# Increasing ratios in finite nonnegative linearization

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**Abstract.** For finite three-term recurrences normalized to have row sums one, the companion paper [5] proves nonnegative linearization (property (P)) when the ratio $\rho(i) = c_i/(b_i + c_i)$ of the downward coefficients to the profile $d_i = b_i + c_i$ is constant and the Newton coefficients of the profile are log-concave without internal zeros. We show that the ratio may increase with the index. The main result: if the Newton coefficients of the profile and those of the ratio are nonnegative and log-concave without internal zeros, and the diagonal coefficients $b_i$ are nondecreasing, then property (P) holds; this admits, for instance, exponential ratios. Separate theorems admit, with the same hypotheses on the profile and on $b$, all convex quadratic ratios and, under the single inequality $r\sigma \le \frac{7}{4}t\tau$, the cubic ratios $r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$. The proofs resum the moves of the kernel recurrence of [5]. The key inequalities of the main result and of the cubic criterion are proved with exact computer certificates, and all results are formalized in Lean 4. Nondecreasing $b$ alone, or nonnegative Newton coefficients of the ratio alone, do not suffice. We also give a self-contained proof, in the Newton basis, of Walkup's theorem that log-concavity of Newton coefficients is preserved under pointwise products.

**Keywords.** Finite orthogonal polynomials; nonnegative linearization; log-concave sequences; Newton series; computer-assisted proof.

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## 1. Introduction

Szwarc [10, §5, Problem 2] asked for criteria, in the recurrence coefficients, for nonnegative linearization (property (P)) of finite systems of orthogonal polynomials. The companion paper [5] answers this for arrays whose diagonal and downward coefficients have a constant ratio: property (P) holds whenever the Newton coefficients of the profile are log-concave without internal zeros. This paper treats ratios $\rho(i) = c_i/(b_i + c_i)$ that increase with $i$.

Our main result admits every ratio whose Newton coefficients are log-concave, on every profile allowed in [5]. Here the Newton coefficients of a sequence $(u_i)$ are $\Delta^k u(0)$, so that $u_n = \sum_k \binom{n}{k}\Delta^k u(0)$.

**Theorem A** *(Theorem 5.5).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients*

*that are log-concave without internal zeros. Let* $\rho(x) = \sum_k \rho_k \binom{x}{k}$*, where* $(\rho_k)_{k\ge 0}$ *is nonnegative and log-concave without internal zeros and* $\rho_0 > 0$*. Put* $a_i = 1 - d_i$*,* $b_i = (1-\rho(i))d_i$ *and* $c_i = \rho(i)d_i$*. If* $b$ *is nondecreasing, then the system has nonnegative linearization, including the products beyond the terminal degree.*

Two examples. On the linear profile $d_i = i/N$, a linear ratio $\rho(i) = r + \kappa i/N$ with $\kappa \ge 0$ has Newton coefficients $(r, \kappa/N)$, so Theorem A applies whenever $b$ is nondecreasing, that is, whenever $\kappa \le \frac{N}{2N-1}(1-r)$. On every admissible profile, the exponential ratios $\rho(x) = r(1+\varepsilon)^x$, whose Newton coefficients $r\varepsilon^k$ are geometric, are admitted whenever $b$ is nondecreasing. For constant $\rho$, Theorem A is the criterion of [5].

Separate results treat cases that Theorem A does not cover. Ratios whose Newton coefficients are not log-concave are admitted when they are convex quadratic, including those with $t^2 < r\tau$ (Theorem 4.2), or cubic with $r\sigma \le \frac{7}{4}t\tau$ (Theorem 5.6 and Corollary 5.7). Profiles whose Newton coefficients are not log-concave are admitted when the Newton coefficients of $c$ are log-concave and those of $b$ are those of $c$ times nonincreasing weights (Theorem 3.1 and Corollary 3.2). For arrays quadratic in the index (Proposition 3.3) and for profiles of Newton degree at most two (Theorem 5.1), weaker conditions suffice. Section 6 summarizes the hierarchy in two tables.

The proofs of Theorem A and of the cubic criteria are computer-assisted. Before the proof of Theorem A, a certificate map states which steps are symbolic certificates and which are finite checks and names the Lean declarations that prove them; the supplement [9] gives the command that reruns each certificate. All theorems, corollaries and propositions, and Lemma 5.4, are formalized in Lean 4.

Section 2 recalls the setting. Section 3 extends the kernel recurrence of [5] to independent diagonal and downward coefficients. Section 4 resums the moves of negative weight for linear and convex quadratic ratios, under the condition that $b$ be nondecreasing, which also appears in the criteria of Askey and Szwarc for infinite systems [10, Theorems 3.1 and 3.2] and in the finite criteria of [6]. Section 5 treats ratios of higher degree, including Theorem A. Section 7 states open problems; Appendix A collects the counterexamples and sampled tests, and Appendix B proves Walkup's product theorem (Proposition 5.3) in the Newton basis.

*Imported and known.* Imported from [5], on which this paper depends, are the finite realization of the arrays [5, Lemma 2.1], the kernel recurrence [5, Lemma 3.2] with the computation in its proof, and the detour inequalities and operator induction [5, Lemma 3.4 and proof of Theorem 3.1], which Section 3 extends; the log-concavity inequality of Lemma 2.1 is also from [5], and its proof is repeated here. Nothing else from [5] is used. Previously known are the constant-ratio case [5], Proposition 5.3, which is Walkup's theorem [12] and is given a self-contained proof in the Newton basis, and the Abel argument of Lemma 5.4, which is that of Wang and Yeh [13].

## 2. Setting

We use the conventions of [5]. An array of real coefficients $(a_i, b_i, c_i)_{0\le i\le N}$ satisfies

$$\begin{aligned} a_0 = 1, \quad b_0 = c_0 = 0, \quad a_N = 0, \quad a_i + b_i + c_i = 1, \\ a_i > 0 \ (i < N), \qquad c_i > 0 \ (i > 0), \end{aligned} \tag{2.1}$$

and $R_{-1} = 0$, $R_0 = 1$ and $xR_i = a_iR_{i+1} + b_iR_i + c_iR_{i-1}$ $(0 \le i < N)$ define its polynomials, with $R_i(1) = 1$. As in [5] we call such arrays *normalized*: their recurrence matrix has row sums one, but (2.1) allows $b_i < 0$, and we call an array *stochastic* only when all $b_i \ge 0$. By [5, Lemma 2.1] the array has a positive orthogonality measure on $N+1$ real points, which include 1, and on this support $R_mR_n = \sum_k g(m,n;k)R_k$ with $g(m,n;k) = [R_m(J)]_{n,k}$, where $J$ is the tridiagonal recurrence matrix; the array has *nonnegative linearization*, or *property (P)*, if all $g(m,n;k) \ge 0$. We write $d_i = b_i + c_i$ for the profile and, for $i \ge 1$ with $d_i > 0$, $\rho(i) = c_i/d_i$ for the ratio.

The Newton coefficients of a sequence $(u_i)_{0\le i\le N}$ are $\Delta^k u(0) = \sum_{i\le k}(-1)^{k-i}\binom{k}{i}u_i$, so that $u_n = \sum_k \binom{n}{k}\Delta^k u(0)$. A nonnegative, finitely supported sequence $(u_k)_{k\ge0}$ is *log-concave without internal zeros* if its positive entries form an interval and $u_k^2 \ge u_{k-1}u_{k+1}$ for all $k \ge 1$. For such a sequence put $f_n(\ell) = \sum_{h\ge0}\binom{\ell}{h}u_{n+h}$, and, for integers $p, q \ge 0$,

$$W_{p,q}(\ell, a) = \sum_{n\ge p} \binom{n-p+q}{q}\binom{a}{n} f_n(\ell),$$

with $A_j = W_{2j+1,j}$, $B_j = W_{2j+2,j}$ and $C_j = W_{2j+2,j+1}$ when these are formed from the Newton coefficients of the profile.

**Lemma 2.1** *([5]). Let $(u_k)_{k\ge0}$ be nonnegative, finitely supported and log-concave without internal zeros. Then $(u_k + u_{k+1})_{k\ge0}$ has the same property, hence so does $(f_n(\ell))_{n\ge0}$ for every $\ell \ge 0$, and*

$$f_i(\ell)\, f_l(\ell+1) \ge f_0(\ell+1)\, f_{i+l}(\ell) \qquad (i, l \ge 1,\ \ell \ge 0).$$

*Proof.* For a nonnegative finitely supported sequence $u$ that is log-concave without internal zeros, $v_i = u_i + u_{i+1}$ has the same property, since $v_i^2 - v_{i-1}v_{i+1} = (u_i^2 - u_{i-1}u_{i+1}) + (u_{i+1}^2 - u_iu_{i+2}) + (u_iu_{i+1} - u_{i-1}u_{i+2})$ and the last term is nonnegative by log-concavity. The sequence $(f_n(\ell))_n$ is obtained from $(u_n)_n$ by applying $u \mapsto v$ $\ell$ times. For such a sequence $g$, balanced products dominate: $g_ig_l \ge g_0g_{i+l}$ and $g_ig_{l+1} \ge g_1g_{i+l}$ for $i, l \ge 1$. Adding these with $g = f(\ell)$ and using $g_l + g_{l+1} = f_l(\ell+1)$ and $g_0 + g_1 = f_0(\ell+1)$ gives the claim. ∎

For the profile, with $u_k = \nu_k$ and $\nu_0 = 0$, one has $f_0(\ell) = d_\ell$, and the inequality reads $f_i(\ell)f_l(\ell+1) \ge d_{\ell+1}f_{i+l}(\ell)$. The paper [5] proves property (P) for constant ratios by expanding the product coefficients in a kernel $\omega_{\ell,a}$ on $\Omega = \{(\ell, a) : \ell, a \ge 0,\ \ell + a \le N\}$, which satisfies a recurrence in $\ell$ whose moves have weights formed from the $W_{p,q}$, and by matching each move of odd index with two-step detours through moves of even index [5, Lemmas 3.2 and 3.4]. We extend that argument.

## 3. Ratios given by weights

The kernel argument also works directly for normalized arrays, with independent diagonal and downward coefficients. Let $a_i + b_i + c_i = 1$ for $0 \le i \le N$, with $b_0 = c_0 = 0$, $a_N = 0$, $a_i > 0$ for $i < N$ and $c_i > 0$ for $i \ge 1$. Write

$$\kappa_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} c_i, \qquad \beta_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} b_i \qquad (1 \le k \le N),$$

with $\kappa_k = \beta_k = 0$ for $k = 0$ and $k > N$, and put $\varphi_n(\ell) = \sum_{h \ge 0} \binom{\ell}{h} \kappa_{n+h}$ and $\psi_n(\ell) = \sum_{h \ge 0} \binom{\ell}{h} \beta_{n+h}$, so that $\varphi_0(\ell) = c_\ell$ and $\psi_0(\ell) = b_\ell$ for $\ell \le N$.

**Theorem 3.1** *(Varying ratio).* *Let $(\kappa_k)$ be log-concave without internal zeros and $\beta_k \ge 0$ for all $k$. Suppose that for $0 \le \ell \le N-1$*

$$\psi_1(\ell)\, \varphi_n(\ell+1) \ge c_{\ell+1}\, \psi_{n+1}(\ell) \quad (n \ge 2), \qquad \varphi_2(\ell)\, \psi_n(\ell+1) \ge c_{\ell+1}\, \psi_{n+2}(\ell) \quad (n \ge 1).$$

*Then the system has nonnegative linearization.*

**Corollary 3.2.** *If $(\kappa_k)$ is log-concave without internal zeros and $\beta_k = w_k \kappa_k$ for $k \ge 1$ with $w_1 \ge w_2 \ge \cdots \ge 0$, then the system has nonnegative linearization. The ratio $c_i/(b_i + c_i)$ is nondecreasing in $i \ge 1$, and it is constant when $w$ is constant on the support of $\kappa$; these are the arrays of [5, Theorem 3.1].*

For example, if $c$ is a convex quadratic in $i$ ($\kappa_1, \kappa_2 > 0$, $\kappa_k = 0$ for $k \ge 3$) and $b_i = w_1 \kappa_1 i + w_2 \kappa_2 \binom{i}{2}$ with $w_1 > w_2 \ge 0$, scaled so that $b_N + c_N = 1$, the ratio $c_i/(b_i + c_i)$ increases strictly. The hypotheses exclude the counterexample recorded in Appendix A, whose diagonal coefficients $b = (0, 1, 1, 1)/91$ have Newton coefficients of both signs. For $N \ge 2$, a strictly increasing linear ratio on a linear profile violates the corollary's assumptions because $\Delta^2 b(0) < 0$; Proposition 3.3 covers it when $b$ is nondecreasing.

*Proof of Theorem 3.1.* Put $e_i = \prod_{j<i} a_j/c_{j+1}$ and $E_i = e_i R_i$. Then $xE_i = c_{i+1}E_{i+1} + b_i E_i + a_{i-1}E_{i-1}$ on the support. The out-of-range terms are omitted: at the endpoints, $xE_0 = c_1 E_1$ and $xE_N = b_N E_N + a_{N-1}E_{N-1}$. The series $F = \sum_i E_i z^i$ satisfies $xF = \mathcal{L}_z F$ with

$$\mathcal{L}_z = z^{-1} c(\theta_z) + b(\theta_z) + z\big(1 - b(\theta_z) - c(\theta_z)\big),$$

where $c(\theta) = \sum_k \kappa_k \binom{\theta}{k}$ and $b(\theta) = \sum_k \beta_k \binom{\theta}{k}$. As in [5, Section 3], $K_k = \sum_{m,n} L^k_{mn} z^m w^n$, with $E_m E_n = \sum_k L^k_{mn} E_k$, satisfies $(\mathcal{L}_z - \mathcal{L}_w) K_k = 0$, and $K_k = \sum \omega_{\ell,a} S^a Z^\ell$ with $\omega_{0,a} = \mathbf{1}_{a=k}$ and $\omega_{\ell,a} = 0$ for $\ell + a > N$. The computation in the proof of [5, Lemma 3.2] gives

$$\frac{(\mathcal{L}_z - \mathcal{L}_w) S^a Z^\ell}{z - w} = S^a Z^\ell + \sum_{p \ge 0} \binom{a}{p} S^{a-p} Z^\ell \Big( \varphi_p(\ell)(\chi_{p-2} - \chi_p) + \psi_p(\ell)(\chi_{p-1} - \chi_p) \Big),$$

with $\chi_{-2} = -Z^{-1}$ and $\chi_{-1} = 0$. Collecting terms as there, with $W_{p,q}$ formed from $\varphi$ and

$W^{\psi}_{p,q}$ from $\psi$, gives

$$\begin{aligned} c_{\ell+1}\omega_{\ell+1,a} \;=\; a_{\ell+a}\,\omega_{\ell,a} + \sum_{j\ge 0}(-1)^j\Big(&A^{\psi}_j(\ell-j,a+2j+1)\,\omega_{\ell-j,a+2j+1} \\ &+ B_j(\ell-j,a+2j+2)\,\omega_{\ell-j,a+2j+2} \\ &+ \big(C_j + C^{\psi}_j\big)(\ell-j-1,a+2j+2)\,\omega_{\ell-j-1,a+2j+2}\Big), \end{aligned}$$

where $A^{\psi}_j = W^{\psi}_{2j+1,j}$, $C^{\psi}_j = W^{\psi}_{2j+2,j+1}$, and $B_j = W_{2j+2,j}$, $C_j = W_{2j+2,j+1}$. The straight weight is $1 - c_{\ell+a} - b_{\ell+a} = a_{\ell+a}$, extended polynomially beyond $N$, and $\beta_0 = 0$. Up to this point no hypothesis of the theorem has been used: the recurrence holds for every normalized array, whatever the signs of its diagonal coefficients. Under the hypotheses, all weights are nonnegative on $\Omega$.

For $j \ge 1$, $0 \le \ell \le N-1$ and $a \ge 0$, the following detour inequalities hold, with weights having a negative second argument interpreted as zero:

$$\begin{aligned} A^{\psi}_0(\ell,a)B_{j-1}(\ell+1,a-1) + B_0(\ell,a)A^{\psi}_{j-1}(\ell+1,a-2) \;&\ge\; 4c_{\ell+1}A^{\psi}_j(\ell,a), \\ B_0(\ell,a)B_{j-1}(\ell+1,a-2) \;&\ge\; 4c_{\ell+1}B_j(\ell,a), \\ B_0(\ell,a)\big(C_{j-1}+C^{\psi}_{j-1}\big)(\ell+1,a-2) \;&\ge\; 4c_{\ell+1}\big(C_j+C^{\psi}_j\big)(\ell,a). \end{aligned}$$

The proof of [5, Lemma 3.4] applies, with the lower bounds $A^{\psi}_0(\ell,a) \ge a\psi_1(\ell)$ and $B_0(\ell,a) \ge \binom{a}{2}\varphi_2(\ell)$. Lemma 2.1 for $\varphi$ handles the products $\varphi_2(\ell)\varphi_n(\ell+1)$. The two hypotheses of the theorem handle $\psi_1(\ell)\varphi_n(\ell+1)$ for $n \ge 2$ and $\varphi_2(\ell)\psi_n(\ell+1)$ for $n \ge 1$, which are the indices that occur. The moves $C_j$ and $C^{\psi}_j$ have the same displacement and are treated as one move. With this, the proof of [5, Theorem 3.1] gives $\omega \ge 0$, hence $L^k_{mn} \ge 0$ and $g(m,n;k) = L^k_{mn}e_k/(e_me_n) \ge 0$. ∎

*Proof of Corollary 3.2.* Set $w_0 = w_1$; since $\kappa_0 = 0$, this leaves $\beta_0 = 0$. Fix $\ell$ and write $G_i = \varphi_i(\ell)$, $H_i = \psi_i(\ell)$ and $u_x = \binom{\ell}{x-i}$. For $i \ge 0$,

$$H_iG_{i+1} - H_{i+1}G_i \;=\; \sum_{x,y\ge i} u_xu_y\kappa_x\kappa_{y+1}(w_x - w_{y+1}).$$

Terms with $x \le y$ are nonnegative, and terms with $x = y+1$ vanish. The map $(x,y) \mapsto (y+1,x-1)$ is a bijection from the terms with $x \ge y+2$ onto those with $i+1 \le x \le y$; the remaining terms have $x = i$. A term and its image sum to $\kappa_x\kappa_{y+1}(w_{y+1} - w_x)(u_{y+1}u_{x-1} - u_xu_y) \ge 0$, because $w$ is nonincreasing and the zero-extended binomial coefficients are log-concave. Hence $H_{i+1}G_i \le H_iG_{i+1}$.

Since the support of $(G_i)$ is an interval and $\beta_k = 0$ where $\kappa_k = 0$, chaining gives $H_{n+1}G_1 \le H_1G_{n+1}$, $H_{n+2}G_n \le H_nG_{n+2}$ and $H_{n+2}G_{n+1} \le H_{n+1}G_{n+2}$. Lemma 2.1 with $(i,l) = (1,n)$ and $(2,n)$ reads $G_1(G_n + G_{n+1}) \ge CG_{n+1}$ and $G_2(G_n + G_{n+1}) \ge CG_{n+2}$, where $C = G_0 + G_1 = c_{\ell+1}$ and $G_n + G_{n+1} = \varphi_n(\ell+1)$. Since $G_1 \ge \kappa_1 > 0$, $H_1(G_n + G_{n+1}) \ge CH_1G_{n+1}/G_1 \ge CH_{n+1}$. If $G_{n+2} > 0$, then $G_2(H_n + H_{n+1}) \ge G_2H_{n+2}(G_n + G_{n+1})/G_{n+2} \ge CH_{n+2}$; if $G_{n+2} = 0$, then $H_{n+2} = 0$. These are the two hypotheses of Theorem 3.1. The case $i = 0$ of the first display, with $\beta_0 = \kappa_0 = 0$, reads $b_{\ell+1}c_\ell \le b_\ell c_{\ell+1}$;

for $1 \le \ell < N$, division by $c_\ell c_{\ell+1} > 0$ shows that $b_i/c_i$ is nonincreasing and $c_i/(b_i + c_i)$ is nondecreasing. If $w$ is constant on the support of $\kappa$, then $b$ is proportional to $c$, and with $d = b + c$ the array is that of [5, Theorem 3.1]. ∎

For arrays that are quadratic in the index, concave diagonal coefficients are allowed.

**Proposition 3.3** *(Quadratic arrays).* *Let $b_n = \beta_1 n + \beta_2 \binom{n}{2}$ and $c_n = \kappa_1 n + \kappa_2 \binom{n}{2}$ $(0 \le n \le N)$ define an array as above. If $\kappa_2 \ge 0$, $\kappa_2 + \beta_2 \ge 0$ and $b_0 \le b_1 \le \cdots \le b_N$, then the system has nonnegative linearization.*

*Proof.* In the recurrence of the proof of Theorem 3.1, which holds for every array, $\varphi_n = \psi_n = 0$ for $n \ge 3$. So all moves with $j \ge 1$ vanish, and the moves with $j = 0$ have the weights $A_0^\psi(\ell, a) = \sum_{n \ge 1} \binom{a}{n} \psi_n(\ell) = b_{\ell+a} - b_\ell$, $B_0(\ell, a) = \binom{a}{2} \kappa_2$ and $(C_0 + C_0^\psi)(\ell, a) = \binom{a}{2}(\kappa_2 + \beta_2)$. These and the straight weights $a_{\ell+a}$ are nonnegative on $\Omega$, so $\omega \ge 0$ by induction on $\ell$, and $g(m, n; k) \ge 0$ as in the proof of Theorem 3.1. ∎

With the linear profile $d_n = n/N$ and the ratio $c_n/d_n = r + \kappa n/N$ $(1 \le n \le N$, $r > 0$, $\kappa \ge 0)$, one has $\kappa_2 = 2\kappa/N^2$ and $\kappa_2 + \beta_2 = 0$ for $N \ge 2$. For $N = 1$, property (P) follows directly from $R_1^2 = c_1 R_0 + b_1 R_1$ when $b_1 \ge 0$. So every linearly increasing ratio with a linear profile and nondecreasing $b$ has property (P); $b$ is nondecreasing exactly when $\kappa \le (1 - r)N/(2N - 1)$. Exact computations [7] show that each of the three hypotheses is needed: dropping any one of them gives quadratic arrays with a negative coefficient at $N \le 3$.

## 4. Linearly and quadratically increasing ratios

For curved profiles, linearly increasing ratios need not satisfy the hypotheses of Theorem 3.1: the Newton coefficients of $b$ of order at least 3 can be negative, and then some moves of even index in the recurrence of its proof have negative weight and no two-step detour. A resummation of the positive moves handles this.

**Theorem 4.1** *(Linearly increasing ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $0 < r \le 1$ and $\kappa \ge 0$, and put $\rho_i = r + \kappa i/N$, $a_i = 1 - d_i$, $b_i = (1 - \rho_i) d_i$ and $c_i = \rho_i d_i$. If $b$ is nondecreasing, then the system has nonnegative linearization. The condition that $b$ be nondecreasing is equivalent to $\kappa\big(N + d_{N-1}/(1 - d_{N-1})\big) \le N(1 - r)$.*

For the linear profile $d_i = i/N$ the condition reads $\kappa \le \frac{N}{2N-1}(1 - r)$, and these are arrays of Proposition 3.3. For a convex profile $d_{N-1} \le (N - 1)/N$, so the admitted range of $\kappa$ contains $[0, \frac{N}{2N-1}(1 - r)]$. When $r < 1$, this containment is strict for a nonlinear profile. For example, $N = 4$, $d_i = (2^i - 1)/15$, $r = 1/2$ and $\kappa = 2/5$ are admitted, with $b = (0, 2/75, 3/50, 7/75, 1/10)$, although $(2N - 1)\kappa/N = 7/10 > 1 - r$.

*Proof.* Write $s = 1 - r$ and $t = \kappa/N$, and let $f_n(\ell) = \Delta^n d(\ell)$ and $\nu_i = \Delta^i d(0)$ be as in Section 2, with $\nu_{-1} = \nu_0 = 0$. For $0 \le i < N$ put $R_i = d_i/f_1(i)$; then $f_1(i) \ge \nu_1 > 0$, and log-concavity of $(f_n(i))_{n \ge 0}$ gives $R_{i+1} - R_i = \big(f_1(i)^2 - d_i f_2(i)\big)/\big(f_1(i) f_1(i+1)\big) \ge$

0. Since $b_{i+1}-b_i = f_1(i)\big(s-t(i+1)-tR_i\big)$ and the bracket is nonincreasing in $i$, $b$ is nondecreasing exactly when its last increment is nonnegative, that is, when $t(N+R_{N-1}) \le s$ with $R_{N-1} = d_{N-1}/(1-d_{N-1})$; this is the stated condition. In particular $tN \le s$. Since $c_i = (r+ti)d_i$ and $b_i = (s-ti)d_i$, the Leibniz rule for differences gives, for $\ell \ge 0$ and $1 \le n \le N-\ell$, which includes every index occurring in the moves on $\Omega$,

$$\varphi_n(\ell) = (r+t(\ell+n))f_n(\ell) + tnf_{n-1}(\ell), \qquad \psi_n(\ell) = h_n(\ell) - v_n(\ell),$$

with $h_n(\ell) = (s-t(\ell+n))f_n(\ell)$ and $v_n(\ell) = tnf_{n-1}(\ell)$; both are nonnegative when $\ell+n \le N$. (Outside this range the formulas need not hold, because the Newton coefficients of $b$ and $c$ are set to zero beyond $N$.) Extend $\nu$ by zero outside $1 \le i \le N$ and put $\widehat{\kappa}_i = p_i\nu_i + q_i\nu_{i-1}$ $(i \ge 1)$, $\widehat{\kappa}_0 = 0$, with $p_i = r+ti$ and $q_i = ti$; the Newton sequence $\kappa$ of $c$ is the truncation of $\widehat{\kappa}$ at $N$. Then

$$\begin{aligned}\widehat{\kappa}_i^2 - \widehat{\kappa}_{i-1}\widehat{\kappa}_{i+1} &= p_i^2(\nu_i^2 - \nu_{i-1}\nu_{i+1}) + q_i^2(\nu_{i-1}^2 - \nu_{i-2}\nu_i)\\ &\quad + q_{i-1}p_{i+1}(\nu_i\nu_{i-1} - \nu_{i-2}\nu_{i+1}) + t^2(\nu_{i-1}\nu_{i+1} + \nu_{i-2}\nu_i + 2\nu_i\nu_{i-1}),\end{aligned}$$

and each term is nonnegative by log-concavity of $\nu$. The positive support of $\widehat{\kappa}$ is that of $\nu$ when $t = 0$ and extends it by one index when $t > 0$, an interval in both cases. Truncation at $N$ preserves log-concavity and interval support, so $\kappa$ is log-concave without internal zeros, and Lemma 2.1 applies to $f$ and to $\varphi$.

*Moves.* We use the recurrence of the proof of Theorem 3.1, which holds for every array. For $j \ge 1$ split $A_j^{\psi} = H_j - V_j$, where $H_j = W^h_{2j+1,j}$ and $V_j = W^v_{2j+1,j}$ are formed from $h_n$ and $v_n$ in place of $\psi_n$; both are nonnegative on $\Omega$. Keep $A_0^{\psi}(\ell,a) = b_{\ell+a} - b_\ell$ as one move; it is nonnegative because $b$ is nondecreasing. The weights $B_j$, formed from $\varphi$, and $C_j + C_j^{\psi} = W^f_{2j+2,j+1}$, formed from $f$, are nonnegative. Write $C'_j = C_j + C_j^{\psi}$.

*Detours.* All detour inequalities below are asserted for source nodes $(\ell,a) \in \Omega$; in the mixed products the occurring indices satisfy $n \le a-2$, hence $\ell+n+2 \le N$. Since $\varphi_2(\ell) \ge (r+t(\ell+1))f_2(\ell)$ and $c_{\ell+1} = (r+t(\ell+1))d_{\ell+1}$, Lemma 2.1 for $f$ gives

$$\begin{gathered}\varphi_2(\ell)f_n(\ell+1) \ge c_{\ell+1}f_{n+2}(\ell), \qquad \varphi_2(\ell)h_n(\ell+1) \ge c_{\ell+1}h_{n+2}(\ell),\\ \varphi_2(\ell)v_n(\ell+1) \ge \tfrac{n}{n+2}\,c_{\ell+1}v_{n+2}(\ell) \quad (n \ge 2),\end{gathered}$$

using $s-t(\ell+n+1) \ge s-t(\ell+n+2) \ge 0$ for the second and Lemma 2.1 with indices $(2,n-1)$ for the third. As in the proof of [5, Lemma 3.4], with $B_0(\ell,a) \ge \binom{a}{2}\varphi_2(\ell)$, for $0 \le \ell \le N-1$ and second factors at $(\ell+1,a-2)$,

$$\begin{aligned} B_0B_{j-1} &\ge 4c_{\ell+1}B_j \ (j \ge 1), & B_0H_{j-1} &\ge 6c_{\ell+1}H_j \ (j \ge 2),\\ B_0V_{j-1} &\ge 6c_{\ell+1}V_j \ (j \ge 2), & B_0C'_{j-1} &\ge 6c_{\ell+1}C'_j \ (j \ge 1).\end{aligned}$$

The first uses Lemma 2.1 for $\varphi$. The coefficient ratios are, on $m \ge 2j+2$, $2j+1$, $2j+1$ and $2j+2$ respectively,

$$\frac{jm(m-1)}{2(m-j-2)}, \qquad \frac{jm(m-1)}{2(m-j-1)}, \qquad \frac{j(m-1)(m-2)}{2(m-j-1)}, \qquad \frac{(j+1)m(m-1)}{2(m-j-1)};$$

they increase with $j$ and are at least 5, 10, 6 and 6. Finally,

$$A_0^{\psi}(\ell,a)B_0(\ell+1,a-1)+B_0(\ell,a)A_0^{\psi}(\ell+1,a-2)\ \ge\ c_{\ell+1}\big(4H_1(\ell,a)-2V_1(\ell,a)\big). \quad (4.1)$$

Only $a\ge 3$ needs proof. Put $u_n=f_n(\ell)$, $y_n=\binom{a}{n}u_n$, $D=u_0+u_1=d_{\ell+1}$, $\alpha=s-t(\ell+a)$,

$$X=\sum_{n\ge1}y_n,\qquad Y=\sum_{n\ge2}y_n,\qquad J=\sum_{n=2}^{a-1}\tbinom{a-1}{n}u_n,$$
$$X'=\sum_{j\ge1}\tbinom{a-2}{j}(u_j+u_{j+1}),\qquad Y'=\sum_{j\ge2}\tbinom{a-1}{j}(u_j+u_{j+1}),$$

so that $X=au_1+Y$, $X'=(a-2)u_1+J$ and $Y'=Y-(a-1)u_2$. Then $A_0^{\psi}(\ell,a)=\alpha X-tau_0$ and $A_0^{\psi}(\ell+1,a-2)=(\alpha+t)X'-t(a-2)D$; both are nonnegative, and the two weights $B_0$ are at least $(r+t(\ell+1))Y'$ and $(r+t(\ell+1))Y$. With $H_*=\sum_{n\ge3}(n-2)y_n$ and $T=\sum_{n\ge2}(n-3)(a-n)y_n$ one has $H_1=\sum_{n\ge3}(n-2)(\alpha+t(a-n))y_n$ and, after a shift of index, $V_1=t\sum_{n\ge2}(n-1)(a-n)y_n$, so $4H_1-2V_1=4\alpha H_*+2tT$. Since $c_{\ell+1}=(r+t(\ell+1))D$, it suffices to show

$$(\alpha X-tau_0)Y'+Y\big((\alpha+t)X'-t(a-2)D\big)\ \ge\ D(4\alpha H_*+2tT). \quad (4.2)$$

The coefficient of $\alpha$ in the difference of the two sides is $K=XY'+YX'-4DH_*$. By Lemma 2.1, $u_i(u_j+u_{j+1})\ge Du_{i+j}$ for $i,j\ge1$, so $XY'\ge au_1Y'\ge D\sum_{n\ge3}ny_n$ and $YX'\ge\binom{a}{2}u_2X'\ge D\sum_{n\ge3}\binom{n}{2}y_n$; hence $K\ge\frac{D}{2}\sum_{n\ge3}\big((n-3)(n-4)+4\big)y_n\ge0$, and (4.2) holds when $t=0$. Let $t>0$. Monotonicity of $b$ at $\ell+a-1$ gives $\alpha\ge tR_{\ell+a-1}\ge tR_\ell=tu_0/u_1$, so it suffices to prove (4.2) at $\alpha=tu_0/u_1$. Both sides are homogeneous of degree two in $(u_n)$; take $u_1=1$ and write $z=u_0$ and $q=u_2$. If $q=0$, then $u_n=0$ for $n\ge2$ and both sides vanish. Otherwise $0\le z\le 1/q$ by log-concavity. At $\alpha=tz$ the two weights $A_0^{\psi}$ are $tzY$ and $t(z+1)J$, and (4.2) divided by $t$ reads $F(z)\ge0$, where

$$\begin{aligned}F(z)\ &=\ zYY'+(z+1)YJ-(z+1)(4zH_*+2T)\\ &=\ (1-qz)F(0)+qzF(1/q)+4H_*z(1/q-z),\end{aligned}$$

the second form holding because $F$ is quadratic with leading coefficient $-4H_*$. It remains to show $F(0)\ge0$ and $F(1/q)\ge0$. Put $L_3=\frac{3}{2}$, $L_n=\frac{n(n+2)}{6}$ for $n\ge4$, and $E_n=\frac{2(n^2-5)}{n+1}$. In $YJ$ keep, for each $n$, the product with indices $(2,2)$ if $n=3$, and the products with indices $(2,n-1)$ and $(3,n-2)$ if $n\ge4$; by log-concavity with $u_1=1$, $u_iu_{n+1-i}\ge u_n$, and $\binom{a-1}{j}\ge\binom{a-i}{j}$, so $YJ\ge\sum_{n=3}^{a-1}L_n(a-n)y_n$. In $Y$ keep the term $\binom{a}{2}q$; as $q(u_{n-1}+u_n)\ge(q+1)u_n$ for $n\ge3$, $YY'\ge(q+1)\sum_{n=3}^{a}\frac{n(a-1)}{2}y_n$. Since $L_n\ge2(n-3)$,

$$F(0)\ =\ YJ-2T\ \ge\ 2(a-2)y_2+\sum_{n=3}^{a-1}\big(L_n-2(n-3)\big)(a-n)y_n\ \ge\ 0.$$

Using $u_n\le qu_{n-1}$ and a shift of index, $\frac{4}{q}H_*+2T\le\sum_{n=2}^{a-1}E_n(a-n)y_n$, where $E_2<0$. As $L_n+\frac{n}{2}\ge E_n$ for $n\ge3$,

$$\frac{F(1/q)}{1+1/q}\ =\ YJ+\frac{YY'}{q+1}-\frac{4}{q}H_*-2T\ \ge\ \sum_{n=3}^{a-1}\Big(L_n+\frac{n}{2}-E_n\Big)(a-n)y_n\ \ge\ 0.$$

The numerical facts used are $L_3 = \frac{3}{2} \geq 0$, $L_3 + \frac{3}{2} = 3 \geq E_3 = 2$, and, for $n \geq 4$, $6(L_n - 2(n-3)) = (n-5)^2 + 11$ and $6(n+1)(L_n + \frac{n}{2} - E_n) = n^3 - 6n^2 + 5n + 60 > 0$.

*Resummation.* Regard the moves as operators on functions on $\Omega$, each divided by $c$ at the destination level; all strictly raise $\ell$, and inequalities between operators are entrywise. Let $P$ and $Q$ be the sums of the $B_j$ with $j$ even and odd, $C = C_+ - C_-$ with $C_\pm$ the sums of the $C'_j$ with $j$ even and odd, $H_+$ and $H_-$ the sums of the $H_j$ with $j \geq 2$ even and $j \geq 3$ odd, $V_+$ and $V_-$ the sums of the $V_j$ with $j$ odd and $j \geq 2$ even, $A_0$ the operator of $A_0^\psi$, and $U$ the straight moves. The recurrence reads $\omega = \beta + (U + P - Q + C + A)\omega$ with $A = A_0 + H_+ - H_- - H_1 + V_+ - V_-$ and $\beta = \omega|_{\ell=0} \geq 0$. The inequalities above give

$$P^2 \geq 4Q, \quad H_+P \geq 6H_-, \quad V_+P \geq 6V_-, \quad C_+P \geq 6C_-, \quad A_0P + PA_0 + 2V_1 \geq 4H_1.$$

Let $G = (I - P/2)^{-1} = \sum_{m=0}^{N} (P/2)^m \geq 0$. Then $PG = GP = 2(G - I)$ and $(I - P/6)G = \frac{2}{3}G + \frac{1}{3}I$, so $G(H_+ - H_-)G \geq GH_+(I - P/6)G \geq 0$, $GCG \geq 0$ and $G(V_+ - V_-)G \geq \frac{2}{3}GV_1G$. Hence

$$\begin{aligned} GAG &\geq G\big(A_0 - H_1 + \tfrac{2}{3}V_1\big)G \geq G\big(A_0 - \tfrac{1}{4}(A_0P + PA_0) + \tfrac{1}{6}V_1\big)G \\ &= \tfrac{1}{2}(GA_0 + A_0G) + \tfrac{1}{6}GV_1G \geq 0. \end{aligned}$$

With $R = U + P^2/4 - Q \geq 0$, the operator $M = G(R + C + A)G$ is nonnegative and strictly raises $\ell$. Since $(I - P/2)^2\omega = \beta + (R + C + A)\omega$, the function $v = (I - P/2)\omega$ satisfies $v = G\beta + Mv$. Therefore $\omega = G\sum_{q=0}^{N} M^q G\beta \geq 0$, and $g(m, n; k) \geq 0$ as in the proof of Theorem 3.1. ∎

**Theorem 4.2** *(Convex quadratic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $\rho(x) = r + tx + \tau\binom{x}{2}$ with $r > 0$ and $t, \tau \geq 0$, and put $a_i = 1 - d_i$, $b_i = (1 - \rho(i))d_i$ and $c_i = \rho(i)d_i$. If $b$ is nondecreasing, then the system has nonnegative linearization.*

This contains Theorem 4.1 ($\tau = 0$). As in the proof of that theorem, $b_{i+1} - b_i = f_1(i)\big(1 - \rho(i+1) - (t + \tau i)R_i\big)$ with a nonincreasing bracket, so $b$ is nondecreasing exactly when $t(N + R) + \tau\big(\binom{N}{2} + (N-1)R\big) \leq 1 - r$ with $R = d_{N-1}/(1 - d_{N-1})$. The Newton coefficients of $c$ need not be log-concave: for $d_i = i/4$ and $\rho(x) = \frac{1}{2} + \binom{x}{2}/40$ they are $(1/8, 1/80, 3/160, 0)$. The proof below replaces their log-concavity by a weighted detour inequality.

*Proof.* Write $f_n(\ell) = \Delta^n d(\ell)$, $\Delta\rho(x) = t + \tau x$ and $R_i = d_i/f_1(i)$, which is nondecreasing as in the proof of Theorem 4.1. Since $b_{i+1} - b_i = f_1(i)\big(1 - \rho(i+1) - \Delta\rho(i)R_i\big)$ and $b$ is nondecreasing, $1 - \rho(m+1) \geq \Delta\rho(m)R_m$ for $0 \leq m < N$, and $\rho(i) \leq 1$ for $i \leq N$. For $\ell + n \leq N$ the Leibniz rule gives, all at $\ell$,

$$\varphi_n = \rho(\ell + n)f_n + n\Delta\rho(\ell + n - 1)f_{n-1} + \binom{n}{2}\tau f_{n-2}, \qquad \psi_n = h_n - v_n - w_n,$$

with $h_n = (1 - \rho(\ell + n))f_n$, $v_n = n\Delta\rho(\ell + n - 1)f_{n-1}$ and $w_n = \binom{n}{2}\tau f_{n-2}$; here $\varphi_n$, $h_n$, $v_n$ and $w_n$ are nonnegative, while $\psi_n$ need not be.

*Moves.* In the recurrence of the proof of Theorem 3.1, keep $A_0^\psi = b_{\ell+a} - b_\ell \ge 0$ and split $A_j^\psi = H_j - V_j - W_j$ for $j \ge 1$, where $H_j$, $V_j$ and $W_j$ are $W_{2j+1,j}$ formed from $h_n$, $v_n$ and $w_n$; the weights $B_j$, formed from $\varphi$, and $C'_j = C_j + C_j^\psi = W^f_{2j+2,j+1}$ are nonnegative. With the signs $(-1)^j$, the moves $W_j$ behave like the $V_j$.

*Detours of $H$, $V$, $W$ and $C'$.* All detours are asserted for sources $(\ell, a) \in \Omega$, with second factors at $(\ell+1, a-2)$, marked by a prime. Since $\varphi_2(\ell) \ge \rho(\ell+1) f_2(\ell)$ and $c_{\ell+1} = \rho(\ell+1) d_{\ell+1}$, Lemma 2.1 for $f$ gives, for $n \le a-2$,

$$\varphi_2 f'_n \ge c_{\ell+1} f_{n+2}, \qquad \varphi_2 h'_n \ge c_{\ell+1} h_{n+2},$$
$$(n+1)(n+2)\varphi_2 v'_n \ge n^2 c_{\ell+1} v_{n+2}, \qquad (n+1)(n+2)\varphi_2 w'_n \ge n(n-1) c_{\ell+1} w_{n+2},$$

the last two for $n \ge 3$. The second uses $1 - \rho(\ell+n+1) \ge 1 - \rho(\ell+n+2) \ge 0$; the third uses Lemma 2.1 with indices $(2, n-1)$ and $(n+1)\Delta\rho(\ell+n) - n\Delta\rho(\ell+n+1) = t + \tau\ell \ge 0$; the fourth uses indices $(2, n-2)$. As in the proof of [5, Lemma 3.4], with $B_0(\ell, a) \ge \binom{a}{2}\varphi_2(\ell)$ and $\binom{a}{2}\binom{a-2}{m-2} = \binom{a}{m}\binom{m}{2}$,

$$B_0 H'_{j-1} \ge 10 c_{\ell+1} H_j, \qquad B_0 V'_{j-1} \ge \tfrac{9}{2} c_{\ell+1} V_j, \qquad B_0 W'_{j-1} \ge 3 c_{\ell+1} W_j \quad (j \ge 2),$$
$$B_0\, C'_{j-1}(\ell+1, a-2) \ge 6 c_{\ell+1} C'_j \quad (j \ge 1).$$

The coefficient ratios are $\frac{jm(m-1)}{2(m-j-1)}$, $\frac{j(m-2)^2}{2(m-j-1)}$ and $\frac{j(m-2)(m-3)}{2(m-j-1)}$ on $m \ge 2j+1$, and $\frac{(j+1)m(m-1)}{2(m-j-1)}$ on $m \ge 2j+2$; they increase with $j$, and at the smallest $j$ the bounds are $(m-5)(m-6) \ge 0$, $(m-5)(2m-7) \ge 0$, $m \ge 5$ and $(m-3)(m-4) \ge 0$.

*Detour of $B$.* We show, for $n \ge 2$ and $\ell + n + 2 \le N$,

$$(n+1)(n+2)\,\varphi_2(\ell)\varphi_n(\ell+1) \ge 8(n-1)\, c_{\ell+1}\varphi_{n+2}(\ell). \tag{4.3}$$

Granting this, $B_0(\ell, a) \ge \binom{a}{2}\varphi_2(\ell)$, (4.3) with $n = m-2$, $\binom{a}{2}\binom{a-2}{m-2} = \binom{a}{m}\binom{m}{2}$ and $\binom{m}{2}\frac{8(m-3)}{m(m-1)} = 4(m-3)$ give

$$\begin{aligned} B_0(\ell, a) B_{j-1}(\ell+1, a-2) &\ge 4c_{\ell+1} \sum_{m \ge 2j+2} (m-3)\binom{m-j-3}{j-1}\binom{a}{m}\varphi_m(\ell) \\ &\ge 4 c_{\ell+1} B_j(\ell, a) \end{aligned}$$

for $j \ge 1$, because $(m-3)\binom{m-j-3}{j-1} - \binom{m-j-2}{j} = \binom{m-j-3}{j-1}\frac{(j-1)(m-2)}{j} \ge 0$. (With constant 1 in place of $\frac{(n+1)(n+2)}{8(n-1)}$, (4.3) can fail: for $N = 20$, $d_i = (2^i - 1)/(2^{20} - 1)$, $r = \frac{1}{2}$, $t = 0$, $\tau = 10^{-3}$ one gets $\varphi_2(16)\varphi_2(17) < c_{17}\varphi_4(16)$.)

To prove (4.3), put $u_k = f_k(\ell)$, so that $f_k(\ell+1) = u_k + u_{k+1}$, and $p = \rho(\ell+1)$, $K_n = (n+1)(n+2)$, $L_n = 8(n-1)$. Both sides are homogeneous of degree two in $u$; take $u_1 = 1$ and write $z = u_0$, $q = u_2$. By convexity and log-concavity, $0 \le z \le \ell$ and $zq \le 1$. If $q = 0$ or $u_n = 0$, then $\varphi_{n+2}(\ell) = 0$. Otherwise put $y = u_{n+1}/u_n \in [0, q]$. Log-concavity gives $u_{n+2} \le y^2 u_n$, and for $n \ge 3$ also $u_{n-1} \ge u_n/q$ and $u_{n-2} \ge u_n/q^2$. Hence $\varphi_n(\ell+1) \ge u_n \mathcal{L}(y)$ and $\varphi_{n+2}(\ell) \le u_n \mathcal{T}(y)$, where

$$\mathcal{T}(y) = \rho(\ell+n+2)y^2 + (n+2)\Delta\rho(\ell+n+1)y + \binom{n+2}{2}\tau,$$

$\mathcal{L}(y) = \rho(\ell+n+1)(1+y) + n\Delta\rho(\ell+n)(1+q^{-1}) + \binom{n}{2}\tau(q^{-1}+q^{-2})$ for $n \ge 3$, and $\mathcal{L}(y) = \rho(\ell+3)(1+y) + 2\Delta\rho(\ell+2)(1+q^{-1}) + \tau(z+1)q^{-1}$ for $n = 2$ (exact). In both cases $\mathcal{L}(y) = A + By$ with $A \ge B \ge 0$, so $\mathcal{L}(0) \ge \mathcal{L}(q)/(1+q)$; and $\mathcal{T}(q) \ge (1+q)\mathcal{T}(0)$ because $(n+2)\Delta\rho(\ell+n+1) \ge (n+2)(n+1)\tau$. The function $\mathcal{F}(y) = K_n\varphi_2(\ell)\mathcal{L}(y) - L_n p(z+1)\mathcal{T}(y)$ is concave with $\mathcal{F}(0) \ge \mathcal{F}(q)/(1+q)$, so it suffices that $\mathcal{F}(q) \ge 0$, which is (4.3) for the configuration $u_0 = z$, $u_k = q^{k-1}$ $(k \ge 1)$.

Next we reduce to $z = \min(\ell, 1/q)$. For $n \ge 3$, only $\varphi_2(\ell) = A_2 + \tau z$ and $c_{\ell+1} = p(z+1)$ depend on $z$, where $A_2 = \rho(\ell+2)q + 2\Delta\rho(\ell+1) \ge \tau$, so $\varphi_2(\ell)/(z+1)$ is nonincreasing in $z$. For $n = 2$, write also $\varphi_2(\ell+1) = G_2 + \tau(z+1)$ with $G_2 = \rho(\ell+3)q(1+q) + 2\Delta\rho(\ell+2)(1+q)$. For $z \le Z \le \ell$ the quotient $\varphi_2(\ell)\varphi_2(\ell+1)/(z+1)$ at $z$ minus its value at $Z$ equals $(Z-z)\big(G_2(A_2-\tau)/((z+1)(Z+1)) - \tau^2\big) \ge 0$, since $G_2(A_2-\tau) \ge 2(\ell+2)(2\ell+1)\tau^2 \ge (\ell+1)^2\tau^2$.

For the remaining configurations we use the following polynomial inequality. For an integer $n \ge 2$ and reals $h \ge 1$, $p, \tau \ge 0$ and $s \ge (h-1)\tau$, put $A = p + (2h-1)s + h^2\tau$, $B = p + nh(s+\tau) + \binom{n}{2}h^2\tau$ and $C = p + ((n+2)h-1)s + \binom{n+2}{2}h^2\tau$. Then

$$K_n AB - L_n pC \ge 0. \tag{4.4}$$

Indeed, $K_nAB - L_npC$ is nondecreasing in $s$ (its $ps$-coefficient is $a_n((n+2)h-1)$ with $a_n = n^2 - 5n + 10 > 0$, and its other $s$-dependent coefficients are nonnegative), and at $s = (h-1)\tau$ it equals $a_np^2 + C_n(h)p\tau + D_n(h)\tau^2$, where $C_n(h) = \frac{1}{2}\big(h^2U_n + h(10n^2+14n-60) + 2a_n\big)$, $U_n = n^4 - 4n^3 - 21n^2 + 12n + 60$ and $D_n(h) = \frac{1}{2}h^2n(n+1)^2(n+2)(3h^2-3h+1)$. For $n \ge 7$ all three coefficients are nonnegative, since $U_n = k^4 + 24k^3 + 189k^2 + 502k + 144$ with $k = n-7$. For $2 \le n \le 6$, $4a_nD_n - C_n^2$ is a polynomial in $v = h-1$ with positive coefficients:

| $n$ | $v^0$ | $v^1$ | $v^2$ | $v^3$ | $v^4$ |
|---|---|---|---|---|---|
| 2 | 576 | 2880 | 5616 | 4992 | 1664 |
| 3 | 1520 | 6240 | 9744 | 7200 | 2160 |
| 4 | 6300 | 27000 | 42660 | 30600 | 8604 |
| 5 | 25200 | 126000 | 229500 | 184800 | 56000 |
| 6 | 62720 | 376320 | 774144 | 677376 | 216576 |

so $a_np^2 + C_np\tau + D_n\tau^2 = a_n\big(p + \frac{C_n\tau}{2a_n}\big)^2 + \frac{4a_nD_n - C_n^2}{4a_n}\tau^2 \ge 0$.

If $1/q \le \ell$, take $z = 1/q$, $h = 1 + 1/q$ and $s = t + \tau\ell \ge (h-1)\tau$. For the configuration above, $\mathcal{F}(q)$ divided by $q(1+q)$ equals $K_nAB - L_npC$, which is nonnegative by (4.4).

If $\ell < 1/q$, take $z = \ell$, and write $w = 1/q - \ell > 0$, $h_0 = \ell + 1$, $h = h_0 + w$ and $s = t + \tau\ell$. With $A(w) = p + (2h-1)s + \tau\big((\ell+2)h - \ell - 1\big)$ and $B(w)$, $C(w)$ as above (with this $h$), $\mathcal{F}(q)$ divided by $q^2$ equals $\mathcal{E}_n(w) = K_n(h_0+w)A(w)B(w) - L_nh_0pC(w)$ for $n \ge 3$, and $\mathcal{E}_2(w) - 12A(w)\tau w(\ell+w)^2$ for $n = 2$. At $w = 0$ both equal $h_0(K_nAB - L_npC)$ with $h = h_0$, which is nonnegative by (4.4). Expanding in $w$, with $\bar{A} = A(0) - p$ and $\bar{B} = B(0) - p$, every coefficient of $w$, $w^2$, $w^3$ and $w^4$ is a sum of nonnegative terms: for $n \ge 3$,

$$\begin{aligned}
[w] &= K_np^2 + p(a_1t + b_1\tau) + K_n\big(\bar{A}\bar{B} + h_0(\eta\bar{B} + \bar{A}\theta)\big),\\
[w^2] &= p(a_2t + b_2\tau) + K_n\big(\eta\bar{B} + \bar{A}\theta + h_0(\eta\theta + \bar{A}\epsilon)\big),\\
[w^3] &= K_n\big(\eta\theta + A(0)\epsilon + h_0\eta\epsilon\big), \qquad [w^4] = K_n\eta\epsilon,
\end{aligned}$$

where

$$\eta = 2s+(\ell+2)\tau, \qquad \theta = n(s+\tau)+n(n-1)\tau h_0, \qquad \epsilon = \tbinom{n}{2}\tau,$$
$$a_1 = (n+2)\big(2\ell(n^2-n+6)+2n^2-3n+11\big), \qquad a_2 = K_n(n+2),$$
$$b_1 = \tfrac{n+2}{2}(\ell^2P_n+\ell Q_n+R_n), \qquad b_2 = \tfrac{K_n}{2}\big(3n(n-3)(\ell+1)+14\ell+12\big),$$

with $P_n = 3k^3+15k^2+6k+8$, $Q_n = 6k^3+30k^2+20k+16$ and $R_n = 3k^3+15k^2+16k+16$ for $k = n-3$; for $n = 2$ the coefficients of $w$, $w^2$ and $w^3$ are

$$12p^2+4(16\ell+13)pt+8(\ell+1)(2\ell+3)p\tau+48(\ell+1)(3\ell+2)t^2$$
$$\quad+12(36\ell^3+92\ell^2+76\ell+21)t\tau+12(27\ell^4+93\ell^3+114\ell^2+62\ell+13)\tau^2,$$
$$48pt+12(2\ell+3)p\tau+24(6\ell+5)t^2$$
$$\quad+12(36\ell^2+67\ell+29)t\tau+12(27\ell^3+78\ell^2+67\ell+19)\tau^2,$$
$$12\big(2s+(\ell+2)\tau\big)\big(2t+(3\ell+5)\tau\big).$$

This proves (4.3).

*Key inequality.* We show, for $(\ell,a)\in\Omega$ with $a\ge 3$,

$$A_0^{\psi}(\ell,a)B_0(\ell+1,a-1)+B_0(\ell,a)A_0^{\psi}(\ell+1,a-2) \ge c_{\ell+1}\big(4H_1-2V_1-W_1\big)(\ell,a). \quad (4.5)$$

Use the notation of the proof of Theorem 4.1 ($u_n = f_n(\ell)$, $y_n$, $D$, $X$, $Y$, $X'$, $Y'$, $J$, $H_*$, $T$), and put $\alpha = 1-\rho(\ell+a)$, $s = \Delta\rho(\ell+a-1)$, $p = \rho(\ell+1)$ and $Z = \sum_{n\ge2}\binom{a-n}{2}y_n$. Then $\alpha \ge sR_{\ell+a-1} \ge su_0/u_1$ and $s\ge\tau(a-1)$. Backward differences give

$$A_0^{\psi}(\ell,a) = \alpha X - sau_0+\tau\tbinom{a}{2}u_0,$$
$$A_0^{\psi}(\ell+1,a-2) = (\alpha+s)X'-s(a-2)D+\tau\tbinom{a-1}{2}D,$$

both nonnegative, and $B_0(\ell,a)\ge pY$, $B_0(\ell+1,a-1)\ge pY'$, $c_{\ell+1} = pD$. Since $1-\rho(\ell+n) = \alpha+(a-n)s-\tau\binom{a-n}{2}$, reindexing gives $4H_1-2V_1 = 4\alpha H_*+2sT+4\tau Z$ and $W_1 = \tau\sum_{n\ge1}n\binom{a-n}{2}y_n$. The argument of the proof of Theorem 4.1 with $s$ in place of $t$, keeping the surplus of the two endpoints ($F(0)\ge 2(a-2)y_2$ and $F(1/q)\ge\frac23(1+1/q)(a-2)y_2$, from the term $n=2$ and $E_2 = -\frac23$), shows

$$(\alpha X-sau_0)Y'+Y\big((\alpha+s)X'-s(a-2)D\big)-D(4\alpha H_*+2sT) \ge \tfrac23 sD(a-2)y_2$$

for $\alpha u_1\ge su_0$; the right side of this bound is linear in $z = u_0/u_1$ and is subtracted before the interpolation. With $s\ge\tau(a-1)$, the difference of the two sides of (4.5) divided by $p$ is therefore at least

$$\tau\Big(\tbinom{a}{2}u_0Y'+D\Big[\tbinom{a-1}{2}y_1+\tfrac{(a-2)(a+11)}{6}y_2+(2a-5)y_3$$
$$+\sum_{n\ge4}\Big(\tbinom{a-1}{2}+(n-4)\tbinom{a-n}{2}\Big)y_n\Big]\Big) \ge 0.$$

*Resummation.* With $W_\pm$ the sums of the $W_j$ with $j$ odd and $j\ge2$ even, the recurrence reads as in the proof of Theorem 4.1 with $A = A_0-H_1+(H_+-H_-)+(V_+-V_-)+$

$(W_+ - W_-)$, and the inequalities above give $P^2 \ge 4Q$, $H_+P \ge 10H_-$, $V_+P \ge \frac{9}{2}V_-$, $W_+P \ge 3W_-$, $C_+P \ge 6C_-$ and $A_0P + PA_0 + 2V_1 + W_1 \ge 4H_1$. Since $(I - P/\mu)G = (1 - \frac{2}{\mu})G + \frac{2}{\mu}I$, one has $G(H_+ - H_-)G \ge 0$, $GCG \ge 0$, $G(V_+ - V_-)G \ge \frac{5}{9}GV_1G$ and $G(W_+ - W_-)G \ge \frac{1}{3}GW_1G$, hence

$$\begin{aligned} GAG &\ge G\big(A_0 - \tfrac{1}{4}(A_0P + PA_0) + \tfrac{1}{18}V_1 + \tfrac{1}{12}W_1\big)G \\ &= \tfrac{1}{2}(GA_0 + A_0G) + \tfrac{1}{18}GV_1G + \tfrac{1}{12}GW_1G \ge 0. \end{aligned}$$

With $U + P^2/4 - Q \ge 0$ this gives $M \ge 0$, and (P) follows as before. ∎

## 5. Ratios of higher degree

For profiles of Newton degree at most two the ratio can have any degree, provided the Newton coefficients of $c$ are nonnegative and log-concave without internal zeros; no key inequality is needed.

**Theorem 5.1** *(Low-degree profiles).* *Let $d_i = \nu_1 i + \nu_2\binom{i}{2}$ with $\nu_1 > 0$, $\nu_2 \ge 0$ and $d_N = 1$, let $c_0 = 0$, and suppose that the Newton coefficients $\kappa_k = \Delta^k c(0)$, $1 \le k \le N$, are log-concave without internal zeros. Put $a_i = 1 - d_i$ and $b_i = d_i - c_i$. If $b$ is nondecreasing, then the system has nonnegative linearization.*

*Proof.* We use the recurrence of the proof of Theorem 3.1, which holds for every array, with $\varphi_n(\ell) = \Delta^n c(\ell)$, $\psi_n(\ell) = \Delta^n b(\ell)$ and $f_n(\ell) = \Delta^n d(\ell)$ for the polynomial extensions. Since $f_n = 0$ for $n \ge 3$, every move $A_j^\psi$ with $j \ge 1$, which involves only $n \ge 2j+1 \ge 3$, equals $-\widetilde{B}_j$ with $\widetilde{B}_j = W_{2j+1,j}$ formed from $\varphi$; the moves $C_j + C_j^\psi = W^f_{2j+2,j+1}$ vanish for $j \ge 1$, and $C_0 + C_0^\psi = \binom{a}{2}\nu_2 \ge 0$. The weight $A_0^\psi(\ell, a) = b_{\ell+a} - b_\ell$ is nonnegative on $\Omega$, as are the straight weights and all $B_j$ and $\widetilde{B}_j$, since $(\varphi_n(\ell))_n$ is obtained from $(\kappa_k)$ by the shifts of Lemma 2.1 and is therefore nonnegative and log-concave without internal zeros. Lemma 2.1 for $\varphi$ gives, as in [5, Lemma 3.4], $B_0(\ell, a)B_{j-1}(\ell+1, a-2) \ge 4c_{\ell+1}B_j(\ell, a)$ for $j \ge 1$, and with $B_0(\ell, a) \ge \binom{a}{2}\varphi_2(\ell)$ also $B_0(\ell, a)\widetilde{B}_{j-1}(\ell+1, a-2) \ge 4c_{\ell+1}\widetilde{B}_j(\ell, a)$ for $j \ge 2$, because the coefficient quotients $jm(m-1)/(2(m-j-1))$ on $m \ge 2j+1$ are at least 10. In the resummation of the proof of Theorem 4.1 take $H = 0$ and $V_j = \widetilde{B}_j$; the key inequality is then trivial, and $M \ge 0$ gives (P). ∎

**Corollary 5.2.** *In Theorem 5.1 let $c_i = \rho(i)d_i$ with $\rho(x) = \sum_k \rho_k\binom{x}{k}$, where $(\rho_k)_{k\ge0}$ is nonnegative and log-concave without internal zeros and $\rho_0 > 0$. Then the Newton coefficients of $c$ are log-concave without internal zeros. Hence every such ratio, of any degree, with nondecreasing $b$ gives nonnegative linearization on these profiles.*

*Proof.* Put $s_k = \rho_k + \rho_{k-1}$ ($\rho_{-1} = 0$), which is log-concave without internal zeros, and $h = \nu_2/2$. The Leibniz rule gives $\kappa_k = ke_k$ for $k \ge 1$, with $e_k = (\nu_1 + h(k-1))s_k + h(k-1)s_{k-1}$. Write $\alpha = \nu_1 + h(k-1)$ and $\beta = h(k-1)$; then

$$\begin{aligned} e_k^2 - e_{k-1}e_{k+1} \;=\; &\alpha^2(s_k^2 - s_{k-1}s_{k+1}) + \beta^2(s_{k-1}^2 - s_{k-2}s_k) \\ &+ (\beta - h)(\alpha + h)(s_{k-1}s_k - s_{k-2}s_{k+1}) \\ &+ h^2(s_{k-1}s_{k+1} + s_{k-2}s_k + 2s_{k-1}s_k), \end{aligned}$$

and every term is nonnegative for $k \geq 2$ by log-concavity of $s$. So $\kappa_k^2 \geq k^2 e_{k-1} e_{k+1} \geq \kappa_{k-1}\kappa_{k+1}$. If $\kappa_k = 0$ with $k \geq 1$, then $s_k = 0$, so $\rho_k = 0$; as $\rho_0 > 0$ and $\rho$ has no internal zeros, $\rho_m = 0$ for $m \geq k$, and $\kappa_{k+1} = 0$. ∎

For example, on the linear profile $d_i = i/N$ the exponential ratios $\rho(x) = r(1+\varepsilon)^x$ with $r > 0$ and $\varepsilon \geq 0$, whose Newton coefficients $r\varepsilon^k$ are geometric, give nonnegative linearization whenever $b$ is nondecreasing. Nonnegativity of the ratio's Newton coefficients alone does not suffice: the cubic counterexample in Appendix A has the linear profile $d_i = i/4$ and the ratio with Newton coefficients $(\frac{19}{20}, 0, 0, \frac{1}{320})$, which are nonnegative and satisfy the log-concavity inequalities but have internal zeros. Theorem 4.2 and Corollary 5.2 are not comparable: the former admits convex quadratic ratios with $t^2 < r\tau$ on every log-concave Newton profile, the latter ratios of every degree on profiles of Newton degree at most two.

**Products of Newton sequences.** The proof of Corollary 5.2 used only that the Newton coefficients of $c = \rho d$ are log-concave. This holds for every profile.

**Proposition 5.3** *(Products).* *Let $(\rho_i)_{i\geq 0}$ and $(\nu_j)_{j\geq 0}$ be nonnegative and log-concave without internal zeros, extended by zero at negative indices, and let $f(x) = \sum_i \rho_i \binom{x}{i}$ and $g(x) = \sum_j \nu_j \binom{x}{j}$ for $x \in \mathbb{Z}_{\geq 0}$, where the sums are finite. Then the Newton coefficients $\kappa_k = \Delta^k(fg)(0)$ of $fg$ are nonnegative and log-concave without internal zeros.*

The proof, which is self-contained and works in the Newton basis, is given in Appendix B.

In the language of random sets: if $S$ and $T$ are independent random subsets of $[n]$ whose laws are invariant under permutations and whose cardinalities are ultra-log-concave of order $n$ in the sense of Liggett [3], then so is $|S \cup T|$, since $\Pr(|S \cup T| = k) = \binom{n}{k}\kappa_k$ for $\Pr(S = A) = \rho_{|A|}$ and $\Pr(T = A) = \nu_{|A|}$. This closure theorem is Walkup's Theorem 4 for two sets [12]: his random sets are conditionally uniform, and their fundamental sequences are $\rho$ and $\nu$. He derives it from his Theorem 2, which proves that the double-binomial convolution $h_k = \sum_i \sum_r \binom{k}{i}\binom{i}{r}\rho_i \nu_{k-r}$, which is $\kappa_k$ after the substitution $j = k - r$, preserves log-concavity; Proposition 5.3 expresses it in the Newton basis, and the proof in Appendix B is self-contained. Hoggar's theorem [2] is the analogue for the power basis. In terms of the linear map $E$ with $E\binom{x}{j} = x^j$, the Newton coefficients of $f$ are the coefficients of $E(f)$, and $E(fg)$ is Wagner's diamond product of $E(f)$ and $E(g)$ [1, (4.3)], [11, Theorem 2.3]; so Proposition 5.3 states that the diamond product preserves nonnegative coefficient sequences that are log-concave without internal zeros. For the numerator polynomial $W(f)$ of the generating function $\sum_{x\geq 0} f(x)t^x$, whose coefficients expand $f$ in another basis, the pointwise product preserves real zeros of $W$ by Wagner [11], ultra-log-concavity of its coefficients by Brändén, Ferroni and Jochemko [1], and their log-concavity without internal zeros by Liu and Mao [4].

**Log-concave ratios on log-concave profiles.** Corollary 5.2 extends to every log-concave Newton profile. Besides Proposition 5.3, the proof uses a sign criterion for quadratic forms in log-concave sequences.

**Lemma 5.4.** *Let $Q(u) = \sum_{0\le i\le j} q_{ij}u_iu_j$ with $q_{0j} \le 0$ for all $j$ and $q_{ij} \ge 0$ for $2 \le i \le j$, and put $N_W(Q) = \sum_{i\le j,\, i+j=W} q_{ij}$. If $N_W(Q) \ge 0$ for every $W$, then $Q(u) \ge 0$ for every nonnegative sequence $(u_n)_{n\ge0}$ that is log-concave without internal zeros.*

*Proof.* Fix $W$ and put $p_i = u_iu_{W-i}$ for $0 \le i \le W/2$ and $p_{-1} = 0$. Log-concavity without internal zeros makes $p$ nondecreasing. With $T_k = \sum_{k\le i\le W/2} q_{i,W-i}$ we get $\sum_{i\le W/2} q_{i,W-i}p_i = \sum_{k\le W/2} T_k(p_k - p_{k-1})$. Here $T_0 = N_W(Q) \ge 0$, $T_1 = N_W(Q) - q_{0W} \ge 0$, and $T_k \ge 0$ for $k \ge 2$. ∎

The weight sums are the coefficients of $Q$ on geometric sequences: $Q(1, r, r^2, \dots) = \sum_W N_W(Q)r^W$. This Abel argument, with the reduction to the full sum when the negative terms come first, is that of Wang and Yeh [13, Lemma 2.1 and Corollary 2.6].

**Theorem 5.5** *(Log-concave ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros. Let $\rho(x) = \sum_k \rho_k \binom{x}{k}$, where $(\rho_k)_{k\ge0}$ is nonnegative and log-concave without internal zeros and $\rho_0 > 0$. Put $a_i = 1 - d_i$, $b_i = (1 - \rho(i))d_i$ and $c_i = \rho(i)d_i$. If $b$ is nondecreasing, then the system has nonnegative linearization.*

This contains Theorem 4.1 and Corollary 5.2. It also contains the case $t^2 \ge r\tau$ of Theorem 4.2. The exponential ratios $\rho(x) = r(1 + \varepsilon)^x$ are admitted on every log-concave Newton profile.

**Certificate map.** Steps (i)–(v) of the proof below, and the corresponding reductions in the proof of Theorem 5.6, hold for every $N$ and use no computer assistance. The remaining obligations, and the Lean declarations that prove them for every $N$ (prefix `Szkraw.` omitted), are these.

- Theorem 5.5, step (vi): the edge bounds, *symbolic* for every window size $a \ge 3$ (`eK_nonneg`, `eL1_nonneg`, `eR_nonneg`, `eR2_nonneg`, `eKpp_nonneg`, `eQs_nonneg`, `eKa_nonneg`, `ePsi_nonneg`, `eQhat_nonneg`).
- Theorem 5.5, step (vii): the windows $3 \le a \le 5$, *finite*, one exact certificate for each case (`win_MM1_3_2`, …, `win_MM1_5_5`, `win_S20_3`, `win_S20_4`).
- Theorem 5.6: the tail sums of (5.5), (5.6) and (5.4), *symbolic* in $\ell$ and $n$ (`cubicB_gen`, `cubicB_two`, …, `cubicB_five`, `cubicB_corner`, `cubicE_gen`, `cubicE_three`, …, `cubicE_six`), and $\Phi_3^{\mathrm{loc}} \ge 0$, *symbolic* in $a$ (`Phi3_nonneg`).
- Corollary 5.7: the modified coefficient in (5.5) for $n \le 5$, *symbolic* in $\ell$; Lean proves it within the declarations for (5.5) above, which are stated under $r\sigma \le \frac{7}{4}t\tau$.

A symbolic certificate has coefficients that are polynomials or rational functions of the parameters and covers the whole range. A worked example after the proof shows how one obligation becomes a polynomial identity checked by Lean. *To check an obligation, see the supplement [9]*: its certificate inventory gives, for each step, the script and the Lean declaration, and its verification procedure gives the command that reruns the script and

the Lean build and axiom audit that check the declarations; the exact cross-checks there recompute coefficients at finitely many sizes.

*Proof. Setup.* Only $\rho_0, \dots, \rho_N$ enter, and truncating $(\rho_k)$ after index $N$ keeps it log-concave without internal zeros, so we may assume it finitely supported. Write $f_n(\ell) = \Delta^n d(\ell)$, $R_i = d_i/f_1(i)$, which is nondecreasing as in the proof of Theorem 4.1, and $P_k(x) = \Delta^k \rho(x) = \sum_h \binom{x}{h} \rho_{k+h}$. As in the proof of Theorem 4.2, $1 - \rho(m+1) \geq \Delta\rho(m) R_m$ for $0 \leq m < N$, and $0 < \rho_0 \leq \rho(i) \leq 1$ for $i \leq N$; $\rho$ is nondecreasing.

For every integer $x \geq 0$, the sequence $(P_k(x))_k$ arises from $(\rho_k)$ by the shifts of Lemma 2.1, so it is log-concave without internal zeros. Since $\rho(x+1)^2 - \rho(x)\rho(x+2) = P_1(x)^2 - P_0(x)P_2(x) \geq 0$, the positive sequence $(\rho(n))_{n \geq 0}$ is log-concave. (The function $x \mapsto \rho(x)$ of a real variable need not be: $\rho(x) = 1 + x/2 + x^2/2$ has $(\log \rho)''(0) > 0$.) Log-concavity in $k$ gives $\rho(y) P_{k+1}(y) \leq P_1(y) P_k(y)$. Combining these facts, for integers $0 \leq x \leq y$ and $k \geq 1$,

$$\rho(x) P_{k-1}(y+1) \leq \rho(x+1) P_{k-1}(y), \qquad \rho(x) P_k(y) \leq \Delta\rho(x) P_{k-1}(y). \tag{5.1}$$

Indeed, log-concavity of $x \mapsto \rho(x)$ gives $\rho(x)\rho(y+1) \leq \rho(x+1)\rho(y)$, equivalently $\rho(x) P_1(y) \leq \Delta\rho(x)\rho(y)$. For the first inequality, $\rho(y) P_{k-1}(y+1) = \rho(y) P_{k-1}(y) + \rho(y) P_k(y) \leq \rho(y+1) P_{k-1}(y)$; multiply by $\rho(x)$ and divide by $\rho(y) > 0$. For the second, $\rho(x)\rho(y) P_k(y) \leq \rho(x) P_1(y) P_{k-1}(y) \leq \Delta\rho(x)\rho(y) P_{k-1}(y)$. By Proposition 5.3, the Newton coefficients of $c = \rho d$ are log-concave without internal zeros, and so is $(\varphi_n(\ell))_n$ for every $\ell$. For $\ell + n \leq N$ the Leibniz rule gives, all at $\ell$,

$$\varphi_n = \sum_{j=0}^{n} \binom{n}{j} P_j(\ell + n - j) f_{n-j}, \qquad \psi_n = h_n - v_n - \widehat{w}_n,$$

with $h_n = (1 - \rho(\ell + n)) f_n$, $v_n = n\Delta\rho(\ell + n - 1) f_{n-1}$ and $\widehat{w}_n = \sum_{j=2}^{n} w_n^{(j)}$, where $w_n^{(j)} = \binom{n}{j} P_j(\ell + n - j) f_{n-j}$. Here $\varphi_n$, $h_n$, $v_n$, $w_n^{(j)}$ and $\widehat{w}_n$ are nonnegative, while $\psi_n$ need not be.

*Moves and detours.* Split the moves as in the proof of Theorem 4.2, with $W_j$ now formed from $\widehat{w}_n$. Lemma 2.1 for $\varphi$ gives $B_0(\ell, a) B_{j-1}(\ell + 1, a - 2) \geq 4 c_{\ell+1} B_j(\ell, a)$ for $j \geq 1$, as in [5, Lemma 3.4]. The detours of $H$ and $C'$, with constants 10 and 6, hold verbatim: they used only $\varphi_2(\ell) \geq \rho(\ell + 1) f_2(\ell)$, Lemma 2.1 for $f$, and $0 \leq \rho \leq 1$ nondecreasing.

For $V$, the term $\rho(\ell + 2) f_2$ of $\varphi_2$, the first inequality of (5.1) with $k = 2$, $x = \ell + 1$, $y = \ell + n$, and Lemma 2.1 for $f$ with indices $(2, n - 1)$ give

$$\varphi_2(\ell) v_n(\ell + 1) \geq \tfrac{n}{n+2} c_{\ell+1} v_{n+2}(\ell).$$

For $\widehat{w}$, expand $\varphi_2(\ell) = \rho(\ell + 2) f_2 + 2\Delta\rho(\ell + 1) f_1 + P_2(\ell) f_0$, and pair its three terms with the components of $\widehat{w}_n(\ell + 1)$:

- $\rho(\ell + 2) f_2$ with component $i \leq n$, by the first inequality of (5.1) and indices $(2, n - i)$, gives at least $\binom{n}{i} \binom{n+2}{i}^{-1} c_{\ell+1} w_{n+2}^{(i)}(\ell)$;
- $2\Delta\rho(\ell + 1) f_1$ with component $i \leq n$, by the second inequality of (5.1) and indices $(1, n - i)$, gives at least $2\binom{n}{i} \binom{n+2}{i+1}^{-1} c_{\ell+1} w_{n+2}^{(i+1)}(\ell)$;

- $P_2(\ell)f_0$ with component $n$ gives at least $c_{\ell+1}w_{n+2}^{(n+2)}(\ell)$, since $P_2(\ell)P_n(\ell+1) \geq (P_0+P_1)(\ell)P_{n+2}(\ell)$ by log-concavity in $k$, and $f_0(\ell+1) = d_{\ell+1}$.

When $n-i=0$ the index condition is trivial. Collecting by target,

$$\varphi_2(\ell)\widehat{w}_n(\ell+1) \geq c_{\ell+1}\sum_{i=2}^{n+2}\theta_{n,i}w_{n+2}^{(i)}(\ell),$$

where $\theta_{n,2} = \binom{n}{2}\binom{n+2}{2}^{-1}$, $\theta_{n,i} = 1-\frac{i(i-1)}{(n+1)(n+2)}$ for $3\leq i\leq n$, $\theta_{n,n+1}=\frac{2}{n+2}$ and $\theta_{n,n+2}=1$.

As in the proof of Theorem 4.2, the coefficient quotients are $\frac{j(m-1)(m-2)}{2(m-j-1)}$ for $V$, and $\theta_{m-2,i}\frac{jm(m-1)}{2(m-j-1)}$ for $\widehat{W}$, on $m\geq 2j+1$, $j\geq 2$. The first is at least 6, by $(m-4)(m-5)\geq 0$ at $j=2$. The second exceeds 2: it is at least $\frac{j(m-2)(m-3)}{2(m-j-1)} \geq m-2$ for $i=2$, at least $\frac{j(2m-3)}{m-j-1}$ for $3\leq i\leq m-2$, equal to $\frac{j(m-1)}{m-j-1}$ for $i=m-1$, and at least 10 for $i=m$. Hence

$$B_0V'_{j-1} \geq 6c_{\ell+1}V_j \qquad \text{and} \qquad B_0W'_{j-1}\geq 2c_{\ell+1}W_j \qquad (j\geq 2).$$

*Resummation.* We prove below, for $(\ell,a)\in\Omega$ with $a\geq 3$,

$$A_0^{\psi}(\ell,a)B_0(\ell+1,a-1)+B_0(\ell,a)A_0^{\psi}(\ell+1,a-2) \geq c_{\ell+1}\big(4H_1-2V_1\big)(\ell,a). \tag{5.2}$$

With the notation of the proof of Theorem 4.2, the inequalities above give $P^2\geq 4Q$, $H_+P\geq 10H_-$, $V_+P\geq 6V_-$, $W_+P\geq 2W_-$, $C_+P\geq 6C_-$ and $A_0P+PA_0+2V_1\geq 4H_1$. Since $(I-P/\mu)G=(1-\frac{2}{\mu})G+\frac{2}{\mu}I$, one has $G(H_+-H_-)G\geq 0$, $GCG\geq 0$, $G(W_+-W_-)G\geq 0$ and $G(V_+-V_-)G\geq \frac{2}{3}GV_1G$, so

$$GAG \geq G\big(A_0-\tfrac{1}{4}(A_0P+PA_0)+\tfrac{1}{6}V_1\big)G = \tfrac{1}{2}(GA_0+A_0G)+\tfrac{1}{6}GV_1G \geq 0,$$

and (P) follows as before.

*Proof of (5.2): notation.* Use the notation of the proof of Theorem 4.1: $u_n=f_n(\ell)$, which is log-concave without internal zeros, $y_n$, $D$, $X$, $X'$, $Y$, $Y'$, $J$, $H_*$. Put also $A=d_{\ell+a}$, $B=d_{\ell+a-1}$, $F=A-B=f_1(\ell+a-1)$, $X''=B-d_\ell$ and $P_k=P_k(\ell)$. Since $c_{\ell+n}=d_{\ell+n}\sum_k\binom{n}{k}P_k$ for $n\leq a$, the following are linear in $P$:

- $B_0(\ell,a)=\sum_k\beta_kP_k$ and $B_0(\ell+1,a-1)=\sum_k\beta'_kP_k$, where

$$\begin{aligned}
&\beta_0 = Y, \quad \beta_1 = a(A-D), \quad \beta_k=\tbinom{a}{k}A\ (k\geq 2),\\
&\beta'_0 = Y', \quad \beta'_1 = aA-2(a-1)d_{\ell+2}+(a-2)D,\\
&\beta'_2 = \tbinom{a}{2}A-(a-1)d_{\ell+2}, \quad \beta'_k=\tbinom{a}{k}A\ (k\geq 3);
\end{aligned}$$

- $c_{\ell+a}-c_\ell=\sum_k\chi_kP_k$ and $c_{\ell+a-1}-c_{\ell+1}=\sum_k\chi'_kP_k$;
- $c_{\ell+a}-c_{\ell+a-1}=\sum_k\delta_kP_k$ with $\delta_k=\binom{a}{k}F+\binom{a-1}{k-1}B$;
- $H_1=H_*-\sum_k\zeta_kP_k$ with $\zeta_k=\sum_{n\geq 3}(n-2)\binom{n}{k}y_n$, and $V_1=\sum_k\nu_kP_k$ with $\nu_k=\sum_{n\geq 3}(n-2)n\binom{n-1}{k-1}\binom{a}{n}u_{n-1}$;

- $c_{\ell+1} = D(P_0 + P_1)$.

As $A_0^{\psi}(\ell, a) = X - (c_{\ell+a} - c_\ell)$ and $A_0^{\psi}(\ell+1, a-2) = X' - (c_{\ell+a-1} - c_{\ell+1})$, the difference of the two sides of (5.2) is $\mathrm{Key}(P) = \mathrm{Lin}(P) - \mathrm{Quad}(P)$. Here $\mathrm{Lin}(P) = \sum_k l_k P_k$ with $l_k = X\beta'_k + \beta_k X' - 4[k \le 1]DH_*$, and Quad is a quadratic form. We use repeatedly:

$$AX'' \ge (a-1)d_\ell F, \qquad AX' \ge (a-2)DF, \qquad u_0 F \le u_1 B, \qquad FD \le u_1 A. \tag{5.3}$$

The first two hold because $A/F \ge R_{\ell+a-1} \ge R_{\ell+i}$ gives $(A/F)f_1(\ell+i) \ge d_{\ell+i}$ for $i \le a-1$; summing over $0 \le i \le a-2$, respectively $1 \le i \le a-2$, gives them. The last two follow from $u_0 u_{m+1} \le u_1 u_m$, as $F = \sum_m \binom{a-1}{m} u_{m+1}$, $B = \sum_m \binom{a-1}{m} u_m$ and $D = u_0 + u_1$.

*(i) Reduction to a quadratic form in $P$.* By monotonicity of $b$ at $\ell + a - 1$, $\Delta c := c_{\ell+a} - c_{\ell+a-1} \le F$. With $\Phi(P) := \Delta c \cdot \mathrm{Lin}(P) - F\,\mathrm{Quad}(P)$ one has $\mathrm{Key} = (1 - \Delta c/F)\mathrm{Lin} + \Phi/F$, so it suffices that every $l_k \ge 0$ and $\Phi \ge 0$. With $\chi_k = \binom{a}{k}A - [k=0]d_\ell$ and $\chi'_k = \binom{a-1}{k}B - \binom{1}{k}D$, so that $E^1_m = X\delta_m - F\chi_m$, $E^2_m = X'\delta_m - F\chi'_m$ and $E^3_m = 2F\nu_m - 4(H_*\delta_m - F\zeta_m)$, expansion gives $\Phi(P) = \sum_{k,m} M_{k,m} P_k P_m$ with

$$M_{k,m} = \beta'_k E^1_m + \beta_k E^2_m + [k \le 1] D\, E^3_m,$$

and explicitly $E^1_0 = E^2_0 = E^3_0 = 0$ and, for $m \ge 1$,

$$E^1_m = \tbinom{a-1}{m-1}AX'' - \tbinom{a-1}{m}d_\ell F, \qquad E^2_m = \tbinom{a-1}{m-1}AX' - \Big(\tbinom{a-1}{m} - [m=1]\Big)DF,$$
$$E^3_m = 2F\nu_m - 4\tbinom{a-1}{m-1}BH_* - 4FZ_m, \qquad Z_m = \sum_{n\ge 3}(n-2)\Big(\tbinom{a}{m} - \tbinom{n}{m}\Big)y_n.$$

Since $(P_k)$ is log-concave without internal zeros, the Abel summation in the proof of Lemma 5.4, applied to every anti-diagonal of $\Phi$, shows $\Phi \ge 0$ if every central sum $S_{\sigma,i} = \sum_{k=i}^{\sigma-i} M_{k,\sigma-k}$, $0 \le i \le \sigma/2$, is nonnegative. As $M_{k,0} = 0$, for $\sigma \ge 3$

$$S_{\sigma,0} = S_{\sigma,1} + M_{0,\sigma} \qquad \text{and} \qquad S_{\sigma,1} = S_{\sigma,2} + M_{1,\sigma-1} + M_{\sigma-1,1}.$$

Since $\binom{a-1}{m} = \frac{a-m}{m}\binom{a-1}{m-1}$ and $\frac{a-m}{m(a-1)} \le \frac{a-m}{m(a-2)} \le \frac{1}{2}$ for $m \ge 2$, (5.3) gives, for $m \ge 2$,

$$E^1_m \ge \tfrac{1}{2}\tbinom{a-1}{m-1}AX'' \ge 0 \qquad \text{and} \qquad E^2_m \ge \tfrac{1}{2}\tbinom{a-1}{m-1}AX' \ge 0.$$

Also every $\beta_k \ge 0$ and $\beta'_k \ge 0$; for $\beta'_2$ use $\binom{a}{2}A \ge (a-1)d_{\ell+2}$. Finally $E^1_1 = AX'' - (a-1)d_\ell F \ge 0$ and $E^2_1 = AX' - (a-2)DF \ge 0$.

*(ii) The central sums with $i \ge 2$, and the coefficients $l_k$.* For $k, m \ge 2$, $M_{k,m} = \beta'_k E^1_m + \beta_k E^2_m \ge 0$, so $S_{\sigma,i} \ge 0$ for $i \ge 2$. Next, $l_0 = K \ge 0$ by the proof of Theorem 4.1, and $l_k \ge 0$ for $k \ge 2$ termwise. By convexity, $A - D \ge (a-1)f_1(\ell+1) = (a-1)(d_{\ell+2} - D)$, so $\beta'_1 \ge (a-2)(A-D)$ and

$$l_1 \ge L := (A-D)\big((a-2)X + aX'\big) - 4DH_*.$$

*(iii) The central sums with $\sigma \le 2$.* Take $P = (P_0, P_1, 0, \ldots)$ with $P_0 \ge 0$ and $P_1 > 0$, and put $\lambda = F/\Delta c(P)$. Then $\mathrm{Key}(\lambda P) = (\lambda/\Delta c)\Phi(P)$, and $\lambda P$ is the linear ratio $\tilde{\rho}(\ell +$

$x) = \lambda(P_0 + P_1 x)$. Its $\tilde{b} = (1-\tilde{\rho})d$ satisfies $\Delta\tilde{b}(\ell + a - 1) = 0$. So $\alpha := 1 - \tilde{\rho}(\ell + a) = tR_{\ell+a-1} \geq tu_0/u_1$ with $t = \lambda P_1$. For such $\alpha$ and $t$, the proof of Theorem 4.1 shows (4.2). Its two weights $A_0^{\psi}$ are nonnegative, since $X \geq au_1$ and $X' \geq (a-2)u_1$, and $B_0(\ell, a) \geq \tilde{\rho}(\ell + 1)Y$, $B_0(\ell+1, a-1) \geq \tilde{\rho}(\ell+1)Y'$, since $\beta_1 \geq Y$ and $\beta_1' \geq Y'$. Hence $\mathrm{Key}(\lambda P) \geq 0$, that is, $\Phi(P) = P_1(M_{0,1}P_0 + M_{1,1}P_1) \geq 0$. This gives $S_{1,0} = M_{0,1} \geq 0$ and $S_{2,1} = M_{1,1} \geq 0$. The sum $S_{2,0} = M_{0,2} + M_{1,1}$ is treated in (v).

*(iii′)* $M_{1,m} + M_{m,1} \geq 0$ *for* $m \geq 2$ *and* $a \geq 6$. Put $c'_m = \binom{a-1}{m-1}$. Then

$$M_{1,m} + M_{m,1} = \big[\beta_1 E^2_m - 4c'_m DBH_*\big] + \big[\beta'_1 E^1_m - DF(4Z_m - 2\nu_m)\big] + \beta'_m E^1_1 + \beta_m E^2_1,$$

and the last two terms are nonnegative.

- First bracket: $E^2_m \geq \frac{1}{2}c'_m AX'$ and $A \geq B$, so it is nonnegative if $R := a(A-D)X' - 8DH_* \geq 0$.
- Second bracket: $Z_m \leq \binom{a}{m}H_* \leq \frac{a}{2}c'_m H_*$, $\nu_m \geq 0$, $F \leq A$, $E^1_m \geq \frac{1}{2}c'_m AX''$ and $\beta'_1 \geq (a-2)(A-D)$. So it is nonnegative if $R_2 := (a-2)(A-D)X'' - 4aDH_* \geq 0$.

*(iv)* $M_{0,s} \geq 0$ *for* $s \geq 3$. Put $c' = \binom{a-1}{s-1}$ and $c'' = \binom{a-1}{s}$. Then

$$M_{0,s} = Y'E^1_s + YE^2_s + DE^3_s = c'BK'' + FG_s,$$

where

$$K'' = Y'X'' + YX' - 4DH_*, \quad G_s = c'(Y'X'' + YX') - c''(d_\ell Y' + DY) + D(2\nu_s - 4Z_s).$$

Note $u_1 = d_{\ell+1} - d_\ell > 0$. If $K'' \geq 0$, then $u_1 B \geq u_0 F$ gives $u_1 M_{0,s} \geq F(c'u_0 K'' + u_1 G_s)$, and

$$c'u_0K'' + u_1G_s = D\,Q_s + c''u_1^2Y',$$
$$Q_s = c'(Y'X'' + YX') - 4c'u_0H_* + u_1(2\nu_s - 4Z_s) - c''u_1(Y + Y').$$

So $M_{0,s} \geq 0$ once $K'' \geq 0$ and $Q_s \geq 0$.

*(v)* $S_{2,0} \geq 0$ *for* $a \geq 5$. With $c'' = \binom{a-1}{2}$, the definitions give the identity

$$S_{2,0} = \beta'_1 E^1_1 + \big[\beta_1 E^2_1 - c''F(d_\ell Y' + DY)\big] + B\,K_a + F\,W,$$

where $K_a = (a-1)K'' - 4DH_*$ and $W = (a-1)(Y'X'' + YX') + D(2\nu_1 + 2\nu_2 - 4Z_1 - 4Z_2)$.

- Since $\beta_1 = a(X' + F)$, $E^2_1 = AX' - (a-2)DF$, $d_\ell \leq D$ and $FD \leq u_1 A$, the bracket is at least $A\Psi$, where $\Psi = a(A-D)J - c''u_1(Y + Y')$. Recall $J = X' - (a-2)u_1$ and $X' + F = A - D$.
- If $K_a \geq 0$, then $u_1(BK_a + FW) \geq F(u_0K_a + u_1W) = FD\widehat{Q}$, where $\widehat{Q} = (a-1)(Y'X'' + YX') - 4au_0H_* + u_1(2\nu_1 + 2\nu_2 - 4Z_1 - 4Z_2)$.

Hence $u_1 S_{2,0} \geq u_1\beta_1' E_1^1 + u_1 A\Psi + FD\widehat{Q} \geq 0$ once $K_a$, $\Psi$ and $\widehat{Q}$ are nonnegative.

*(vi) Edge bounds.* Steps (ii)–(v) reduce (5.2) to the nonnegativity of the quadratic forms $K$, $L$, $R$, $R_2$, $K''$, $Q_s$ $(3 \leq s \leq a)$, $K_a$, $\Psi$ and $\widehat{Q}$ in the sequence $u$. For $Q_s$, first remove the dependence on $s$: by $\nu_s \geq 0$, $Z_s \leq \binom{a}{s} H_*$, $\binom{a}{s} \leq \frac{a}{3}\binom{a-1}{s-1}$ and $\binom{a-1}{s} \leq \frac{a-3}{3}\binom{a-1}{s-1}$ for $s \geq 3$, one has $Q_s \geq \binom{a-1}{s-1} Q^*$ with

$$Q^* \;=\; Y'X'' + YX' - 4u_0 H_* - \tfrac{4a}{3} u_1 H_* - \tfrac{a-3}{3} u_1 (Y + Y').$$

Each of these forms is a sum of products $xy$ of linear forms $x = \sum_{i\geq 1} x_i u_i$ and $y = \sum_{j\geq 1} y_j u_j$ with nonnegative coefficients, plus $u_1$ times a linear form, minus $u_0$ times a linear form with nonnegative coefficients. Since $u$ is log-concave without internal zeros, $u_i u_j \geq u_1 u_{i+j-1}$ for $i, j \geq 1$ and $u_0 u_{n+1} \leq u_1 u_n$. Keeping in $xy$ only the pairs with $i \leq 2$ or $j \leq 2$ gives

$$xy \;\geq\; u_1 \sum_n e_n(x,y) u_n, \qquad e_n(x,y) \;=\; x_1 y_n + x_2 y_{n-1} + [n \geq 3]\, y_1 x_n + [n \geq 4]\, y_2 x_{n-1},$$

and $u_0 \sum_{n\geq 1} h_n u_n \leq u_1 \sum_{n\geq 0} h_{n+1} u_n$. So each form is at least $u_1 \sum_n e_n u_n$ with explicit coefficients $e_n$, and it suffices that every $e_n \geq 0$.

- For $4 \leq n \leq a$ write $n = 4 + p$ and $a = n + k$. Then $e_n / \binom{a}{n}$ is a rational function of $(p, k)$ whose numerator has nonnegative coefficients and whose denominator is strictly positive for $p, k \geq 0$.

- For $n \leq 3$, $e_n$ is a polynomial in $a$ whose coefficients are nonnegative after $a \mapsto a + a_0$, with $a_0 = 6$ for $R_2$, $a_0 = 5$ for $K_a$ and $a_0 = 3$ otherwise.

- For $n \geq a + 1$ only nonnegative products remain.

These certificates are formalized in Lean, and `runs/lean_plan/edge_exact.py` checks every $e_n$ in exact arithmetic for $a \leq 60$.

*(vii) Small windows.* Steps (iii′) and (v) need $a \geq 6$ and $a \geq 5$. For $3 \leq a \leq 5$, each sum $M_{1,m} + M_{m,1}$ $(2 \leq m \leq a)$ and, for $a \leq 4$, $S_{2,0}$ is a cubic form in $u$ that is an exact nonnegative rational combination of the products $u_k(u_{i+1}u_{j-1} - u_i u_j)$ $(i + 1 \leq j - 1)$ and of monomials (`runs/lean_plan/window_cert.py` in [7]). This proves (5.2) and the theorem. ∎

**A worked certificate.** The smallest obligation shows how a step of the proof becomes a polynomial identity checked by Lean. In step (vii) with $a = 3$, the central sum $S_{2,0} = M_{0,2} + M_{1,1}$ must be nonnegative. Substituting the definitions of step (i), with $u_n = f_n(\ell)$, gives a cubic form in $u_0, \ldots, u_3$ with integer coefficients,

$$\begin{aligned} S_{2,0} \;=\; & 24u_1^3 + 135u_1^2u_2 + 189u_1u_2^2 + 66u_2^3 + 21u_1^2u_3 + 98u_1u_2u_3 + 52u_2^2u_3 \\ & + 11u_1u_3^2 + 10u_2u_3^2 + 12u_0u_1u_2 + 12u_0u_2^2 - 42u_0u_1u_3 - 29u_0u_2u_3 \\ & - 11u_0u_3^2 - 12u_0^2u_3, \end{aligned}$$

and the obligation is $S_{2,0} \geq 0$ for every nonnegative sequence $u$ that is log-concave without internal zeros. For such $u$ the products $u_k(u_{i+1}u_{j-1} - u_iu_j)$ with $i+1 \leq j-1$ are nonnegative, and so are all monomials. The script `runs/lean_plan/window_cert.py` solves an exact linear program for nonnegative rational multipliers of these generators and returns

$$\begin{aligned} S_{2,0} \;=\; & 36u_0(u_1u_2 - u_0u_3) + 42u_0(u_2^2 - u_1u_3) + 24u_1(u_1^2 - u_0u_2) + 189u_1(u_2^2 - u_1u_3) \\ & + 135u_2(u_1^2 - u_0u_2) + 29u_3(u_1^2 - u_0u_2) + 98u_3(u_1u_2 - u_0u_3) \\ & + 24u_0^2u_3 + 105u_0u_2^2 + 87u_0u_3^2 + 181u_1^2u_3 + 11u_1u_3^2 + 66u_2^3 + 52u_2^2u_3 + 10u_2u_3^2, \end{aligned}$$

an identity of polynomials, which the script checks by expansion. The Lean statement `Szkraw.win_S20_3` asserts $S_{2,0} \geq 0$ for every such sequence. Its proof records the fifteen generators above as nonnegative facts, expands $S_{2,0}$ from the Lean definitions of the $M_{k,m}$, and closes the goal by linear arithmetic, which the Lean kernel checks. The symbolic certificates of step (vi) use the transfers $u_iu_j \geq u_1u_{i+j-1}$ and $u_0u_{n+1} \leq u_1u_n$ in place of these generators, and their coefficients $e_n$ are rational functions of $a$ and $n$ whose numerators have nonnegative coefficients, so one certificate covers every window size.

For cubic ratios, log-concavity of the Newton coefficients can be replaced by a single inequality. We state the bound $r\sigma \leq t\tau$ first because it follows from log-concavity of $(r,t,\tau,\sigma)$ and its written proof needs only tail-sum certificates; the stronger bound $r\sigma \leq \frac{7}{4}t\tau$ of Corollary 5.7 needs further certificates for small $n$. The Lean proof establishes the corollary directly and derives the theorem from it.

**Theorem 5.6** *(Cubic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, and let $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$ with $r > 0$, $t,\tau,\sigma \geq 0$ and $r\sigma \leq t\tau$. Put $a_i = 1 - d_i$, $b_i = (1-\rho(i))d_i$ and $c_i = \rho(i)d_i$. If $b$ is nondecreasing, then the system has nonnegative linearization.*

If $(r,t,\tau,\sigma)$ is log-concave without internal zeros, then $r\sigma \leq t\tau$: for $t\tau > 0$ this follows from $t^2\tau^2 \geq r\tau \cdot t\sigma$, and otherwise $\sigma = 0$. Theorem 5.6 also admits cubic ratios with $t^2 < r\tau$ or $\tau^2 < t\sigma$, which Theorem 5.5 does not cover. The failing cubic examples of Appendix A violate $r\sigma \leq t\tau$. The proof is computer-assisted: the inequalities (5.5), (5.6) and (5.4) and the inequality $\Phi_3^{\mathrm{loc}} \geq 0$ below are reduced to families of explicit polynomials with nonnegative coefficients.

*Proof. Setup.* As in the proof of Theorem 4.2, $R_i = d_i/f_1(i)$ is nondecreasing, $1 - \rho(m+1) \geq \Delta\rho(m)R_m$ for $0 \leq m < N$, and $\rho$ is nondecreasing with $0 < \rho(i) \leq 1$ for $i \leq N$; here $\Delta\rho(x) = t + \tau x + \sigma\binom{x}{2}$ and $\Delta^2\rho(x) = \tau + \sigma x$. For $\ell + n \leq N$ the Leibniz rule gives, all at $\ell$,

$$\varphi_n \;=\; \rho(\ell+n)f_n + n\Delta\rho(\ell+n-1)f_{n-1} + \tbinom{n}{2}\Delta^2\rho(\ell+n-2)f_{n-2} + \tbinom{n}{3}\sigma f_{n-3}.$$

Since $\binom{\ell+n-1}{2} = \binom{\ell}{2} + \ell(n-1) + \binom{n-1}{2}$ and $n\binom{n-1}{2} = \binom{n}{2}(n-2) = 3\binom{n}{3}$, this gives $\psi_n = h_n - v_n - e_n$ with $h_n = (1-\rho(\ell+n))f_n$,

$$v_n \;=\; n\Gamma_n(\ell)f_{n-1}, \qquad \Gamma_n(\ell) \;=\; t + \tau(\ell+n-1) + \sigma\Big(\tbinom{\ell}{2} + \ell(n-1)\Big),$$
$$e_n \;=\; \tbinom{n}{2}(\tau + \sigma\ell)f_{n-2} + \sigma\tbinom{n}{3}\big(3f_{n-1} + 3f_{n-2} + f_{n-3}\big),$$

all nonnegative. Write $u_k = f_k(\ell)$, so that $f_k(\ell+1) = u_k + u_{k+1}$; then $(u_k)$ is log-concave, and $u_0 \le \ell u_1$ by convexity.

*Moves.* Split the moves as in the proof of Theorem 4.2, with $V_j$ formed from $v_n$ and with a third family $E_j = W_{2j+1,j}$ formed from $e_n$ in place of $W_j$.

*Detours of $H$, $V$ and $C'$.* The detours of $H$ and $C'$ hold verbatim, with constants 10 and 6. For $V$,

$$(n+1)\Gamma_n(\ell+1) - n\Gamma_{n+2}(\ell) = t + \tau\ell + \sigma\Big(\tbinom{\ell}{2} + n^2 - 1\Big) \ge 0,$$

so Lemma 2.1 with indices $(2, n-1)$ gives $(n+1)(n+2)\varphi_2 v_n' \ge n^2 c_{\ell+1} v_{n+2}$ for $n \ge 3$, and $B_0 V_{j-1}' \ge \frac{9}{2} c_{\ell+1} V_j$ for $j \ge 2$ as before. (With the full $\sigma$-part $n\sigma\binom{\ell+n-1}{2} f_{n-1}$ in $v_n$, the corresponding difference is negative for $\ell < n+1$; without any $\sigma$-part in $v_n$, the reduction of the key inequality below fails for large $\ell$.)

*Detour of $E$.* We show, for $n \ge 3$ and $\ell + n + 2 \le N$,

$$(n+1)(n+2)\Big(\varphi_2(\ell) + \frac{n}{3}\varphi_3(\ell)\Big) e_n(\ell+1) \ge 3(n-1)\, c_{\ell+1} e_{n+2}(\ell). \tag{5.4}$$

Granting this, $B_0(\ell, a) \ge \binom{a}{2}\varphi_2(\ell) + \binom{a}{3}\varphi_3(\ell)$, $\binom{a}{3}\binom{a-2}{n} = \frac{a-2}{3}\binom{a}{2}\binom{a-2}{n}$ with $a - 2 \ge n$ whenever $\binom{a}{n+2} \ne 0$, and $\binom{a}{2}\binom{a-2}{n} = \binom{a}{n+2}\binom{n+2}{2}$ give, for $j \ge 2$,

$$B_0(\ell, a) E_{j-1}(\ell+1, a-2) \ge \tfrac{3}{2} c_{\ell+1} \sum_{m \ge 2j+1} (m-3)\tbinom{m-j-2}{j-1}\tbinom{a}{m} e_m(\ell) \ge 3 c_{\ell+1} E_j(\ell, a),$$

because $(m-3)\binom{m-j-2}{j-1} - 2\binom{m-j-1}{j} = \binom{m-j-2}{j-1}\frac{(j-2)(m-1)}{j} \ge 0$.

*Detour of $B$.* We show, for $n \ge 2$, $\ell + n + 2 \le N$ and $(\ell, n) \ne (0, 2)$,

$$(n+1)(n+2)\,\varphi_2(\ell)\varphi_n(\ell+1) \ge 8(n-1)\, c_{\ell+1}\varphi_{n+2}(\ell), \tag{5.5}$$

and, if $N \ge 4$,

$$\big(12\varphi_2(0) + 8\varphi_3(0)\big)\varphi_2(1) \ge 8 c_1 \varphi_4(0). \tag{5.6}$$

Granting these, the argument after (4.3) gives $B_0(\ell, a) B_{j-1}(\ell+1, a-2) \ge 4 c_{\ell+1} B_j(\ell, a)$ for $j \ge 1$. The only exception is the term $m = 4$ for $\ell = 0$, $j = 1$. There $B_0(0, a) \ge \binom{a}{2}\varphi_2(0) + \binom{a}{3}\varphi_3(0)$, $\binom{a}{2}\binom{a-2}{2} = 6\binom{a}{4}$ and $\binom{a}{3}\binom{a-2}{2} = 2(a-2)\binom{a}{4} \ge 4\binom{a}{4}$ reduce the term to (5.6). Inequality (5.5) fails at $(\ell, n) = (0, 2)$ in general.

*Proof of (5.5), (5.6) and (5.4).* In each, the difference of the two sides is a quadratic form $\sum \kappa_{\beta\gamma}\rho_\beta\rho_\gamma$ in $(\rho_0, \rho_1, \rho_2, \rho_3) = (r, t, \tau, \sigma)$. Its coefficients are bilinear forms in the $u_k$, with coefficients polynomial in $\ell$ and $n$.

- Since $t\tau\kappa_{t\tau} + r\sigma\kappa_{r\sigma} = (t\tau - r\sigma)\kappa_{t\tau} + r\sigma(\kappa_{t\tau} + \kappa_{r\sigma})$ and $r\sigma \le t\tau$, the form is nonnegative if $\kappa_{t\tau}$, $\kappa_{t\tau} + \kappa_{r\sigma}$ and all coefficients other than $\kappa_{r\sigma}$ are nonnegative as forms in $u$.

- For (5.5) the coefficients $\kappa_{rr}$, $\kappa_{r\tau}$, $\kappa_{\tau\tau}$ need not be checked: the form with $t = \sigma = 0$ is (4.3) for the ratio $r + \tau\binom{x}{2}$. The same holds for (5.6), whose left side exceeds that of (5.5) at $(0, 2)$ by $8\varphi_3(0)\varphi_2(1) \ge 0$.

- Every remaining coefficient is a sum $\sum c_{ij}u_iu_j$ over $i \le j$. Group its terms by $i+j = m$. Within a group, $u_iu_{m-i}$ is nondecreasing in $i$ by log-concavity, so with the tail sums $T_k = \sum_{i\ge k} c_{i,m-i}$ and the smallest index $k_0$ of the group,

$$\sum_i c_{i,m-i}u_iu_{m-i} \;=\; T_{k_0}u_{k_0}u_{m-k_0} + \sum_{k>k_0} T_k\,(u_ku_{m-k} - u_{k-1}u_{m-k+1}),$$

  and the coefficient is nonnegative if all $T_k \ge 0$.

- The tail sums are polynomials in $(\ell, n)$. For (5.5) with $n \ge 6$ and for (5.4) with $n \ge 7$ they have nonnegative coefficients in $(\ell, n-6)$, respectively $(\ell, n-7)$. For each $n$ in $3 \le n \le 5$ ((5.5)) and $3 \le n \le 6$ ((5.4)) they are polynomials in $\ell$ with nonnegative coefficients. The same holds for (5.5) with $n = 2$ and $\ell \ge 1$ after $\ell \mapsto \ell+1$, and for (5.6), where $u_0 = 0$.

The scripts `runs/lean_plan/gen_cubicB.py` and `runs/lean_plan/gen_cubicE.py` check every tail-coefficient sign in exact rational arithmetic and generate the Lean certificates; Lean checks the certificate identities.

*Key inequality.* We show, for $(\ell, a) \in \Omega$ with $a \ge 3$,

$$A_0^{\psi}(\ell,a)B_0(\ell+1,a-1) + B_0(\ell,a)A_0^{\psi}(\ell+1,a-2) \;\ge\; c_{\ell+1}\big(4H_1 - 2V_1 - E_1\big)(\ell,a). \quad (5.7)$$

Use the notation of the proof of Theorem 4.2, and put $P_1 = \Delta\rho(\ell)$ and $P_2 = \Delta^2\rho(\ell) = \tau + \sigma\ell$, so that $\rho(\ell+k) = \rho(\ell) + kP_1 + \binom{k}{2}P_2 + \binom{k}{3}\sigma$. Put $\alpha = 1 - \rho(\ell+a)$, $s_q = P_1 + (a-1)P_2$ and $s = \Delta\rho(\ell+a-1) = s_q + \binom{a-1}{2}\sigma$. Then $\alpha \ge su_0/u_1$ and $s_q \ge (a-1)P_2$. Backward differences give

$$\begin{aligned} A_0^{\psi}(\ell,a) \;&=\; \alpha X - s_q a u_0 + P_2\tbinom{a}{2}u_0 - \sigma\tbinom{a}{3}u_0, \\ A_0^{\psi}(\ell+1,a-2) \;&=\; (\alpha+s)X' - s_q(a-2)D + P_2\tbinom{a-1}{2}D - \sigma\tbinom{a-1}{3}D, \end{aligned}$$

and, reindexing as before,

$$4H_1 - 2V_1 - E_1 \;=\; 4\alpha H_* + 2s_qT + 4P_2Z - P_2\sum_{n\ge1} n\tbinom{a-n}{2}y_n + \sigma(4T_3 - E_3),$$

where $T_3 = \sum_{n\ge3}(n-2)\big(\binom{a}{3} - \binom{n}{3}\big)y_n$ and $E_3 = \sum_{n\ge3}(n-2)\binom{n}{3}\binom{a}{n}(3u_{n-1} + 3u_{n-2} + u_{n-3})$. With $B_0(\ell,a) \ge pY$, $B_0(\ell+1,a-1) \ge pY'$ and $c_{\ell+1} = pD$, it suffices that $R = A_0^{\psi}(\ell,a)Y' + YA_0^{\psi}(\ell+1,a-2) - D(4H_1 - 2V_1 - E_1) \ge 0$. $R$ is affine in $\alpha$, with coefficient $XY' + YX' - 4DH_*$. Its part without $\sigma$ is the expression bounded in the proof of Theorem 4.2, with $s_q$ and $P_2$ in place of $s$ and $\tau$; that bound holds for every value $\alpha'$ of $\alpha$ with $\alpha' u_1 \ge s_q u_0$. Apply it with $\alpha' = \alpha - \binom{a-1}{2}\sigma z$, where $z = u_0/u_1$. This gives $R \ge \sigma\Phi_3^{\mathrm{loc}}$ with

$$\Phi_3^{\mathrm{loc}} \;=\; \tbinom{a-1}{2}z\,(XY' + YX' - 4DH_*) - \tbinom{a}{3}u_0Y' + Y\Big(\tbinom{a-1}{2}X' - \tbinom{a-1}{3}D\Big) - D(4T_3 - E_3).$$

$\Phi_3^{\mathrm{loc}} \ge 0$. Write $E_3 = \binom{a}{3}u_0 + E_3^0$ and $H_*^+ = \sum_{n\ge2}(n-1)\binom{a}{n+1}u_n$, so that $u_0H_* \le u_1H_*^+$. Bounding the only negative term quadratic in $u_0$, $-4\binom{a-1}{2}u_0^2H_*$, this way and

dropping $\binom{a}{3}u_0^2u_1 \geq 0$ gives $u_1\Phi_3^{\mathrm{loc}} \geq u_1A + u_0B$ with

$$A = \binom{a-1}{2}YX' - u_1\Big(\binom{a-1}{3}Y + 4T_3 - E_3^0\Big),$$
$$B = \binom{a-1}{2}\big(XY' + YX' - 4u_1H_* - 4u_1H_*^+\big) - u_1\Big(\binom{a}{3}Y' + \binom{a-1}{3}Y + 4T_3 - E_3^0 - \binom{a}{3}u_1\Big).$$

These are forms of the kind treated in step (vi) of the proof of Theorem 5.5: the edge bounds give $A \geq u_1\sum_n e_n^A u_n$ and $B \geq u_1\sum_n e_n^B u_n$. Their coefficients are nonnegative by the same three kinds of certificates: for $4 \leq n \leq a$ the numerators in $(p,k)$ have nonnegative coefficients and the denominators are strictly positive for $p,k \geq 0$, for $n \leq 3$ the polynomials in $a$ have nonnegative coefficients after $a \mapsto a+3$, and for $n \geq a+1$ only nonnegative terms remain (`runs/lean_plan/phi3_generic.py` and `runs/lean_plan/phi3_edge.py` in [7]). Hence $R \geq 0$.

*Resummation.* Proceed as in the proof of Theorem 4.2, with $E$ in place of $W$. The inequalities $P^2 \geq 4Q$, $H_+P \geq 10H_-$, $V_+P \geq \frac{9}{2}V_-$, $E_+P \geq 3E_-$, $C_+P \geq 6C_-$ and $A_0P + PA_0 + 2V_1 + E_1 \geq 4H_1$ give $GAG \geq \frac{1}{2}(GA_0 + A_0G) + \frac{1}{18}GV_1G + \frac{1}{12}GE_1G \geq 0$, and (P) follows. ∎

**Corollary 5.7** *(Extended cubic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, and let $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$ with $r > 0$, $t,\tau,\sigma \geq 0$ and $r\sigma \leq \frac{7}{4}t\tau$. Put $a_i = 1 - d_i$, $b_i = (1-\rho(i))d_i$ and $c_i = \rho(i)d_i$. If $b$ is nondecreasing, then the system has nonnegative linearization.*

*Proof.* The proof of Theorem 5.6 uses $r\sigma \leq t\tau$ only in (5.5), (5.6) and (5.4), and there only through $\kappa_{t\tau} + \kappa_{r\sigma} \geq 0$; the other coefficients, the key inequality (5.7) and the resummation do not involve the condition. Since

$$t\tau\,\kappa_{t\tau} + r\sigma\,\kappa_{r\sigma} = \big(t\tau - \tfrac{4}{7}r\sigma\big)\kappa_{t\tau} + \tfrac{4}{7}r\sigma\big(\kappa_{t\tau} + \tfrac{7}{4}\kappa_{r\sigma}\big),$$

under $r\sigma \leq \frac{7}{4}t\tau$ it suffices that $\kappa_{t\tau} \geq 0$ and $\kappa_{t\tau} + \frac{7}{4}\kappa_{r\sigma} \geq 0$ as forms in $u$. For (5.4), for (5.5) with $n \geq 6$ and for (5.6), where $u_0 = 0$, the tail sums of the modified coefficient $\kappa_{t\tau} + \frac{7}{4}\kappa_{r\sigma}$ stay nonnegative. For (5.5) with $n \leq 5$ the transfers alone do not suffice. There the modified coefficient is certified after the substitution $u_0 = zu_1$, $0 \leq z \leq \min(\ell, 1/q)$ with $q = u_2/u_1$, in the regimes $z = 0$, $z = \ell$ and $z = 1/q$ (`runs/cubic/bdetour_u.py` in [7]; `runs/cubic/edetour_u.py` certifies (5.4) independently in the same way). The relevant expressions are affine in $z$, so these boundary regimes cover the admissible interval for $q > 0$. When $q = 0$, all $u_k$ with $k \geq 2$ vanish, and the inequality follows directly. ∎

The Lean proof, `Szkraw.cubic_ratio74_matrix`, proves Corollary 5.7 directly and derives Theorem 5.6 from it. For (5.5) with $n \leq 5$ it does not substitute $u_0 = zu_1$: it writes the modified coefficient, times a positive polynomial in $\ell$, as a combination of transfers, products $u_au_b$ and the terms $u_j(\ell u_1 - u_0) \geq 0$, with multipliers that are polynomials in $\ell$ with nonnegative coefficients after a shift of $\ell$, plus exact rational certificates at the few smaller values of $\ell$ (`runs/lean_plan/cert74.py` and `runs/lean_plan/gen_cubicB.py` in [7]; see the supplement [9]). Exact diagnostics for both routes are listed in the documentation [8].

*Remark* 5.8. The factor $\frac{7}{4}$ is a limit of the certificates, not of property (P). On geometric profiles at $(\ell, n) = (0, 3)$ one has $\kappa_{r\sigma} = -20q(4q^3 + 11q^2 + 9q + 1)$ and $\kappa_{t\tau} = 20(q+1)(8q^3 + 33q^2 + 30q + 6)$ in (5.5), so $\kappa_{t\tau}/|\kappa_{r\sigma}| \to 2$ as $q \to \infty$. This family therefore obstructs the coefficientwise test for $r\sigma \le u\,t\tau$ with $u > 2$; it is not a counterexample to property (P). With $u = \frac{9}{5}$ the certificate of `runs/cubic/bdetour_u.py` fails. Whether $r\sigma \le u\,t\tau$ suffices for some $u > \frac{7}{4}$ is open (Section 7).

## 6. Summary of the criteria

Table 1 lists the criteria of this paper with their hypotheses and the earlier criterion that each extends, and Table 2 gives for each an array or class that it covers and the criteria it extends do not.

Table 1: The criteria of this paper and the earlier criterion that each extends. Each array satisfies (2.1), with profile $d_i = b_i + c_i$ and ratio $\rho(i) = c_i/d_i$. "$u$ LC" means that the Newton coefficients of $u$ are nonnegative and log-concave without internal zeros, and "$b\uparrow$" that $b$ is nondecreasing.

| Result | Hypotheses | Extends |
|---|---|---|
| [5] | $d$ LC; $\rho \equiv r \in (0, 1]$ | — |
| Thm. 3.1, Cor. 3.2 | $c$ LC; Newton coefficients of $b$ are $w_k\kappa_k$ with $w_1 \ge w_2 \ge \cdots \ge 0$ | [5] ($w$ constant) |
| Prop. 3.3 | $b$, $c$ quadratic in $i$; $\kappa_2 \ge 0$, $\kappa_2 + \beta_2 \ge 0$; $b\uparrow$ | Cor. 3.2 on quadratic arrays |
| Thm. 4.1 | $d$ LC; $\rho(i) = r + \kappa\frac{i}{N}$, $\kappa \ge 0$; $b\uparrow$ | Prop. 3.3 (linear $d$) |
| Thm. 4.2 | $d$ LC; $\rho = r + tx + \tau\binom{x}{2}$, $t, \tau \ge 0$; $b\uparrow$ | Thm. 4.1 ($\tau = 0$) |
| Thm. 5.1, Cor. 5.2 | $d = \nu_1 i + \nu_2\binom{i}{2}$; $c$ LC (Cor.: $\rho$ LC, any degree); $b\uparrow$ | Prop. 3.3 on these profiles |
| Thm. 5.5 (Theorem A) | $d$ LC; $\rho$ LC, $\rho(0) > 0$; $b\uparrow$ | Thm. 4.1, Cor. 5.2, Thm. 4.2 for $t^2 \ge r\tau$ |
| Thm. 5.6, Cor. 5.7 | $d$ LC; $\rho = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$, $t, \tau, \sigma \ge 0$, $r\sigma \le t\tau$ (Cor.: $r\sigma \le \frac{7}{4}t\tau$); $b\uparrow$ | Thm. 4.2 ($\sigma = 0$), Thm. 5.5 for cubic $\rho$ |

## 7. Open problems

The preceding criteria cover several increasing ratios, but simple shape restrictions alone do not imply property (P). Exact counterexamples show that log-concavity of the Newton coefficients of both $d$ and $c$ with nondecreasing $b$, a merely increasing ratio with nondecreasing $b$, and nonnegative Newton coefficients of a cubic ratio can each fail. The parameters and negative product coefficients are collected in Appendix A, separately from sampled tests, which establish no threshold.

Find direct inequalities for other varying ratios, especially those outside Theorems 3.1, 4.1 and 5.5. Determine whether the cubic bound of Corollary 5.7 can be

Table 2: For each criterion of Table 1, an array or class that it covers and the criteria it extends do not.

| Result | Newly covered |
|---|---|
| [5] | $K_{N,r}$ and $G_{N,q,r}$ |
| Thm. 3.1, Cor. 3.2 | $c$ convex quadratic, $b_i = w_1\kappa_1 i + w_2\kappa_2\binom{i}{2}$ with $w_1 > w_2 \geq 0$ |
| Prop. 3.3 | $d_i = \frac{i}{N}$, $\rho(i) = r + \kappa\frac{i}{N}$ with $0 < \kappa \leq \frac{N(1-r)}{2N-1}$, $N \geq 2$ |
| Thm. 4.1 | $N = 4$, $d_i = \frac{2^i-1}{15}$, $r = \frac{1}{2}$, $\kappa = \frac{2}{5}$ |
| Thm. 4.2 | $d_i = \frac{i}{4}$, $\rho = \frac{1}{2} + \frac{1}{40}\binom{x}{2}$, whose $c$ is not LC |
| Thm. 5.1, Cor. 5.2 | $d_i = \frac{i}{N}$, $\rho = r(1+\varepsilon)^x$ |
| Thm. 5.5 | $\rho = r(1+\varepsilon)^x$ on every LC profile $d$ |
| Thm. 5.6, Cor. 5.7 | cubic $\rho$ with $\sigma > 0$ and $t^2 < r\tau$ or $\tau^2 < t\sigma$ (Cor.: also $t\tau < r\sigma \leq \frac{7}{4}t\tau$) |

enlarged, and whether the log-concavity assumption on ratios of degree greater than three can be weakened to conditions like the one in Theorem 5.6. The coefficientwise detour test fails on geometric profiles beyond the factor 2 (Remark 5.8), which is a limit of that test rather than a counterexample to property (P). Find a proof of the key inequality (5.2) that does not use computer certificates. All these questions include the terminal row and products whose degree sum exceeds $N$.

## A. Exact counterexamples and sampled tests

All examples in this appendix are reproduced in the companion code [7]. The counterexamples of the first paragraph are exact and machine-verified. The tests of the second paragraph are sampled numerical evidence; they are not formalized.

**Exact counterexamples.** Each array below satisfies (2.1) and has a negative product coefficient.

- *Log-concave Newton coefficients of $d$ and $c$ with nondecreasing $b$.* The arrays $d = (0, 25, 55, 91)/91$ and $c = (0, 24, 54, 90)/91$ give $g(3, 3; 3) = -3/2002$.
- *An increasing concave ratio with nondecreasing $b$.* The linear profile $d_i = i/3$ with $\rho = (1/2, 3/4, 5/6)$, for which $b_i = 1/6$ for $i \geq 1$, gives $g(3, 3; 3) = -1/24$.
- *A cubic ratio with nonnegative Newton coefficients.* For $N = 4$, $d_i = i/4$ and $\rho(x) = \frac{19}{20} + \binom{x}{3}/320$, whose Newton coefficients $(\frac{19}{20}, 0, 0, \frac{1}{320})$ are nonnegative but have internal zeros, $b = (0, \frac{1}{80}, \frac{1}{40}, \frac{9}{256}, \frac{3}{80})$ is nondecreasing and $g(4, 4; 4) = -14011/30720000$. So Theorem 4.2 does not extend to cubic ratios without further conditions.
- *Cubic ratios with $r \leq \frac{9}{10}$.* On linear profiles with $\kappa = 1/80$, which keeps $b$ nondecreasing, $\rho_i = \frac{9}{10} + \kappa\binom{i}{3}/\binom{N}{3}$ gives $g(48, 48; 48) < 0$ for $N = 48$, and $\rho_i = \frac{9}{10} + \kappa(i/N)^3$ gives $g(64, 64; 63) < 0$ for $N = 64$.

- *Steep ratios.* On linear profiles, with nondecreasing $b$ and nonnegative Newton coefficients of $\rho$, the ratio $\rho_i = \frac{1}{2} + 10^{i-N}/100$ gives $g(40, 40; 40) < 0$ for $N = 40$. Raising only the last ratio, $\rho_i = r$ for $i < N$ and $\rho_N = r + (1-r)/N$, gives $g(40, 40; 38) < 0$ for $r = \frac{1}{2}$, $N = 40$, and $g(120, 120; 119) < 0$ for $r = \frac{2}{5}$, $N = 120$; these two ratios are convex, with a single kink.

Write a cubic ratio as $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$. The cubic examples violate $r\sigma \le t\tau$, a consequence of log-concavity without internal zeros of $(r, t, \tau, \sigma)$: in the first two $t = \tau = 0$, and for $\kappa(i/N)^3$ one has $t\tau = 6\kappa^2/N^6 < r\sigma = 6r\kappa/N^3$. One further exact computation concerns the method rather than property (P): for $t = \tau = 0$ the operator $M$ of the proof of Theorem 4.1 can have negative entries, for instance for $N = 6$, $d_i = i/6$, $r = \frac{1}{5}$ and $\sigma = \frac{1}{200}$. This obstructs extending that resummation to such data, and does not contradict Theorem 4.1, which does not cover them.

**Sampled tests without failures.** These are numerical evidence over finitely many arrays, not certified thresholds. In sampled arrays with $t, \tau \ge 0$, $r\sigma = u\,t\tau$ and nondecreasing $b$ on log-concave Newton profiles, no negative coefficients were found for $u \in \{\frac{1}{4}, 1, 4, 16, 64\}$, $r \in \{\frac{1}{10}, \frac{1}{3}, \frac{1}{2}, \frac{4}{5}, \frac{19}{20}, \frac{99}{100}\}$ and $4 \le N \le 18$ (full tables), or in the coefficients $g(m, N; k)$ for $u \in \{1, 4, 16\}$ and $N = 24, 32, 48, 64, 96$. In these samples the operator $M$ of the proof of Theorem 4.1 was nonnegative for $u \le 4$ and $N \le 13$. The samples include values $u > \frac{7}{4}$, so they bear on the question of Section 7 whether the bound of Corollary 5.7 can be enlarged, but they do not decide it.

## B. Proof of Proposition 5.3

We recall the statement: if $(\rho_i)_{i\ge0}$ and $(\nu_j)_{j\ge0}$ are nonnegative and log-concave without internal zeros, $f(x) = \sum_i \rho_i\binom{x}{i}$ and $g(x) = \sum_j \nu_j\binom{x}{j}$, then the Newton coefficients $\kappa_k = \Delta^k(fg)(0)$ of $fg$ are nonnegative and log-concave without internal zeros.

*Proof.* The product rule $\binom{x}{i}\binom{x}{j} = \sum_k \binom{k}{i}\binom{i}{k-j}\binom{x}{k}$ gives $\kappa_k = \sum_p H_k(p)\rho_{p_1}\nu_{p_2}$, where $H_k(i,j) = \binom{k}{i}\binom{i}{k-j} = [y^i w^j](y + w + yw)^k$ vanishes unless $i, j \le k \le i + j$. If either sequence vanishes identically, so does $\kappa$. Otherwise, if $\rho$ and $\nu$ are supported on $[p_0, p_1]$ and $[q_0, q_1]$, with $p_1, q_1 \le \infty$, then $\kappa_k > 0$ exactly for $\max(p_0, q_0) \le k \le p_1 + q_1$, so $\kappa$ has no internal zeros. For log-concavity fix $k \ge 1$; then

$$\kappa_k^2 - \kappa_{k-1}\kappa_{k+1} = \sum_{c\in\mathbb{Z}^2}\sum_{p\in\mathbb{Z}^2} T_c(p)\,u_c(p_1)\,w_c(p_2),$$
$$T_c(p) = H_k(p)H_k(c-p) - H_{k-1}(p)H_{k+1}(c-p),$$

with $u_c(i) = \rho_i\rho_{c_1-i}$ and $w_c(j) = \nu_j\nu_{c_2-j}$. By log-concavity without internal zeros, $u_c$ is symmetric about $c_1/2$ and nondecreasing on $i \le c_1/2$, so $u_c = \sum_{0\le a\le c_1/2}(u_c(a) - u_c(a-1))\mathbf{1}_{[a,c_1-a]}$ with nonnegative coefficients, and likewise $w_c$. It therefore suffices to show $D_c(B) = \sum_{p\in B} T_c(p) \ge 0$ for every box $B = [a, c_1 - a] \times [b, c_2 - b]$ with $a \le c_1/2$, $b \le c_2/2$.

Let $E = \{(1,0),(0,1),(1,1)\}$, so that $H_k = \sum_{e\in E} H_{k-1}(\cdot - e)$. Expanding $H_k(p)$, $H_k(c-p)$ and $H_{k+1}(c-p)$ in this way gives

$$D_c(B) = \sum_{e,e'\in E} \big(V_{e+e'}(B-e) - V_{e+e'}(B)\big), \qquad V_s(p) = H_{k-1}(p)H_{k-1}(c-s-p),$$

where $V_s(X) = \sum_{p\in X} V_s(p)$. Fix $s = e+e'$, write $V = V_s$, $m = c-s$, $t = k-1$ and $\mu = m/2$. With $n_1 = 2t-m_2$, $n_2 = 2t-m_1$, $n_3 = m_1+m_2-2t$ one has $V(x,y) = \frac{t!^2}{n_1!n_2!n_3!}\binom{n_1}{t-y}\binom{n_2}{t-x}\binom{n_3}{x+y-t}$ (and $V=0$ if some $n_i<0$). Three properties follow.

(T1) For every interval $W$, the marginal $G_W(x) = \sum_{y\in W} V(x,y)$ is a constant times $\binom{n_2}{t-x}\sum_{z\in t-W}\binom{n_1}{z}\binom{n_3}{x-z}$, a product of a binomial row and a convolution of two log-concave sequences without internal zeros; so $G_W$ is log-concave without internal zeros. The same holds with the roles of the coordinates exchanged.

(T2) $V(c-s-p) = V(p)$, and $V$ is reverse regular of order two: $V(x,y)V(x',y') \le V(x,y')V(x',y)$ for $x\le x'$, $y\le y'$, by log-concavity of $\binom{n_3}{\cdot}$. Taking $x' = 2\mu_1 - x$ and $y,y'$ symmetric about $\mu_2$, central symmetry turns this into $V(x,y)^2 \le V(x,y')^2$. Hence $V(x,y') \ge V(x,y)$ whenever $x\le\mu_1$, $y\le y'$ and $y+y' = 2\mu_2$; and symmetrically in the coordinates.

(T3) If $G$ is log-concave without internal zeros and $G(x) \le G(2\mu_1 - x)$ for all $x\le\mu_1$, then $G$ is nondecreasing on $x\le\mu_1$: otherwise $G(x-1) > G(x)$ for some $x\le\mu_1$, and log-concavity makes $G$ nonincreasing from $x-1$ on, so $G(2\mu_1 - x + 1) \le G(x) < G(x-1)$.

Write $I = [a, c_1 - a]$, $J = [b, c_2 - b]$, $X(I',J') = V(I'\times J')$, and telescope along rows or columns. By the symmetry $H_k(i,j) = H_k(j,i)$ it suffices to treat five ordered pairs.

$e = e' = (1,0)$: $\mu = c/2 - (1,0)$ and $V(B-e) - V(B) = G_J(a-1) - G_J(a-2)$ by central symmetry. Since $J$ is symmetric about $\mu_2$, $G_J$ is symmetric about $\mu_1$, hence nondecreasing on $x\le\mu_1$ by (T1) and (T3); and $a-1\le\mu_1$.

$e = (1,0)$, $e' = (0,1)$: $\mu = (c-(1,1))/2$ and $V(B-e) - V(B) = V(a-1, c_2-b) - V(a-1, b-1) \ge 0$ by (T2), since $a-1 < \mu_1$ and $(c_2-b)+(b-1) = 2\mu_2$.

$e = (1,0)$, $e' = (1,1)$: $\mu = c/2 - (1, \frac12)$ and

$$V(B-e) - V(B) = \big[V(a-1,c_2-b) - V(a-1,b-1)\big] + \big[G_{J-1}(a-1) - G_{J-1}(a-2)\big].$$

The first bracket is nonnegative by (T2), as $a-1\le\mu_1$. For the second, $G_{J-1}(2\mu_1 - x) - G_{J-1}(x) = V(x,c_2-b) - V(x,b-1) \ge 0$ for $x\le\mu_1$ by (T2), so (T1) and (T3) apply.

$e = (1,1)$, $e' = (1,0)$: the same $V$, and $V(B-e) - V(B) = \big[X(I-1,J-1) - X(I-1,J)\big] + \big[V(B-(1,0)) - V(B)\big]$. The first bracket vanishes by central symmetry, since $I-1$ is symmetric about $\mu_1$; the second is the previous case.

$e = e' = (1,1)$: $\mu = c/2 - (1,1)$ and

$$\begin{aligned} V(B-e) - V(B) = {} & \big[K(b-1) - K(b-2)\big] + \big[X(a-1,J) - X(a-1,J-2)\big] \\ & + \big[G_{J-2}(a-1) - G_{J-2}(a-2)\big] \end{aligned}$$

with $K(y) = \sum_{x \in I-1} V(x,y)$. Since $I-1$ is symmetric about $\mu_1$, $K$ is symmetric about $\mu_2$ and nondecreasing on $y \le \mu_2$ by (T1) and (T3); and $b-1 \le \mu_2$. The second bracket is $[V(a-1,c_2-b)-V(a-1,b-2)]+[V(a-1,c_2-b-1)-V(a-1,b-1)]$, two differences of the form in (T2), since both pairs of second coordinates sum to $2\mu_2 = c_2-2$. For the third, $G_{J-2}(2\mu_1-x)-G_{J-2}(x) = X(x,J)-X(x,J-2) \ge 0$ for $x \le \mu_1$ by (T2), so (T1) and (T3) give $G_{J-2}(a-1) \ge G_{J-2}(a-2)$.

Each of the nine ordered-step contributions $V_{e+e'}(B-e)-V_{e+e'}(B)$ is nonnegative, so $D_c(B) \ge 0$, which proves the proposition. (The individual $T_c(p)$ need not be nonnegative.) ■

Diagnostics in the companion code [7] check in exact arithmetic the identity for $D_c(B)$, the sign of each of its nine terms for $k \le 9$, and every bracket of the case analysis, (T1) and (T2) for $k \le 6$.

## AI-use disclosure

Claude Opus 5.5 and, for parts of the Lean formalization of Theorem 5.5, Claude Haiku (Anthropic; the Haiku version was not recorded), both run in the Claude Code agent; GPT-6 Astra or GPT-6.1 Sol (OpenAI; which of the two was not recorded), run in the Codex agent; and GPT-6 Pro (OpenAI), used in ChatGPT, were used to develop the proofs, the exact checks and certificates, the Lean formalization and the text. GPT-6 Pro also reviewed every result. This AI review is not human peer review, and no independent human verification is claimed. The author set the research objective and is responsible for the mathematical claims. The supplement [9] gives the full disclosure, including which results arose with which model.

## Code and formalization

All theorems, corollaries and propositions, and Lemma 5.4, are formalized in Lean 4 with Mathlib in the companion code [7], whose axiom audit admits only `propext`, `Classical.choice` and `Quot.sound`; "machine-verified" refers only to these Lean theorems. The certificate scripts were also run unmodified from a fresh clone at the cited revision [8]. The supplement [9] lists the Lean declaration for each result and remark, gives the inventory of computer-assisted steps with their ranges, scripts and Lean declarations, and lists what is not formalized, chiefly the sampled tests of Appendix A and the comparisons with the literature.

## References

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